

A gasdermin-based life–death evolution system for reprogramming protease specificity

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Reprogramming the specificity of proteases toward alternative target sequences could enable an array of exciting applications, ranging from proteome editing to therapeutic interventions. Here we report an *in vivo* life–death selection system for protease reprogramming using the toxic N-terminal domain of gasdermin D (GD-N) protein as a selection marker. The approach is a modular system that can be used to cover the protease mutational diversity in the billions through only a few cycles of directed evolution. By inserting the desired cleavage sequence into the loop region of the GD-N protein, which is toxic to host bacteria cells, the system selects for efficient substrate cleavage—rendering GD-N nontoxic—by enrichment of bacteria in liquid culture. Using the tobacco etch virus protease (TEVp) and corresponding substrate sequence as a model, we demonstrated that our platform could select and enrich an efficient protease variant millionfold after a single round of selection. We also evolve TEVp to cut sequences on target proteins with known pathological roles.

The precise cleavage of a specific sequence by proteases is fundamental for regulating the functions, localizations and interactions of proteins and for creating new bioactive molecules by cutting zymogens or precursor proteins^{1,2}. Protease activity is crucial for numerous physiological and pathological processes, such as programmed cell death³, the blood coagulation cascade⁴ and pathogen infection⁵. Programmable and sequence-specific cleavage enzymes such as Cas9 are cornerstones for gene editing, making possible almost unlimited possibilities for cutting, changing or reorganizing the genome; therefore, reprogramming the specificity of proteases to cut a new target protein could enable an array of exciting applications, ranging from proteome editing to therapeutic interventions^{6–9}. For example, reprogrammed proteases could allow knocking down or out a target protein, on-demand activation of a desired signal transduction cascade and the elimination of pathological proteins^{9–13}.

Restriction endonucleases and ligases have long been used for the cleavage and ligation of DNA. However, they are not considered gene-editing tools because these enzymes recognize specific DNA sequences that are only 6–8 bases long¹⁴ and such specificity is insufficient for selective targeting across the billions of base pairs in an entire human genome (Fig. 1a). In contrast, the human proteome consists of approximately 20,000 different protein-encoding genes and the length of all protein sequences in the proteome combined is roughly 10 million, indicating that target sequences of 5–7 aa are usually sufficient to achieve specificity in the proteome landscape¹⁵. Interestingly, most proteases bind to their substrates by recognizing from four to eight sequential amino acids¹ (Fig. 1a), implying that, by harnessing the evolutionary potential to refine the substrate recognition specificity of proteases, it may be possible to achieve selective cleavage at specific

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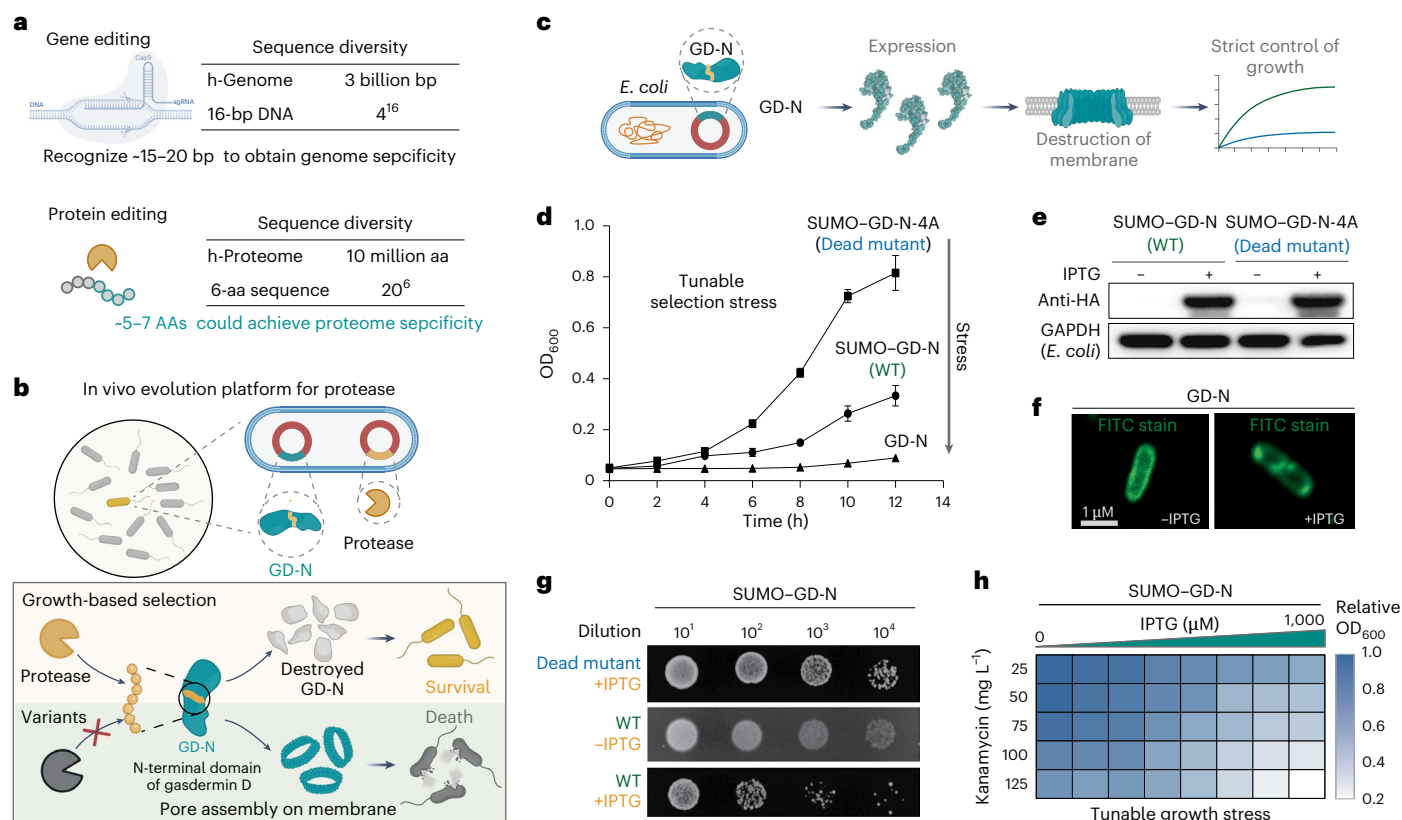


Fig. 1 | A GD-based life–death selection system for protease reprogramming. **a**, The sequence diversity that must be covered to tailor protease specificity through gene or protein editing. The diversity of genomes can be up to 3 billion bases; therefore, if we wish to produce locus-specific gene editing on the entire genome, we need to recognize a length up to 15 bases to be able to do so. This is because the diversity of a 15-mer nucleic acid is 4^{15} , which is matched to the diversity of human genome. However, for protein editing, there are about total 20,000 kinds of proteins and the diversity is much lower than for the genome. We only need to recognize a length of more than 6 aa (20^6 diversity) to be able to accomplish target-specific or sequence-specific editing on the entire proteome. **b**, To construct the in vivo selection system, the target sequence to be recognized by the protease was inserted into the loop domain of the GD-N, a toxic protein. When active, GD-N forms multimeric aggregates that punch holes in the cell membrane, leading to cell death. A protease variant that can cut the target sequence results in cleavage and inactivation of GD-N, allowing the host cell to

survive. **c**, Schematic of the GD-controlled culturing system. **d**, Expression of the GD-N and SUMO-GD-N protein provided sufficient and tunable stress for the growth of *E. coli* in liquid cultures. SUMO-GD-N-4A is a dead mutant of GD-N harboring four substitutions: R137A, K145A, R151A and R153A. **e**, The expression of SUMO-GD-N and SUMO-GD-N-4A could be well controlled using IPTG induction in *E. coli*. **f**, Membrane structure staining of *E. coli* before or after the expression of GD-N. The membrane of *E. coli* was stained with FITC for 1 h and then subjected to laser confocal imaging. **g**, Expression of SUMO-GD-N protein provided sufficient and tunable stress for the growth of *E. coli* in agar. This indicated that the system can be further used for the agar plate selections. **h**, Heat map showing the OD₆₀₀ values of *E. coli* after culturing for 12 h with different concentrations of kanamycin and IPTG demonstrated that *E. coli* growth could be modulated by different concentrations of antibiotic and by GD-N expression level. Schematic diagrams were created with BioRender.com.

target sites within the entire proteome. Such an approach could effectively enable the editing of proteins, analogous to how nucleases are used to edit DNA sequences within the genome (Fig. 1a).

To specifically target and cut a unique sequence on a protein of interest, it is essential to precisely alter the recognition sequence of a protease. Therefore, reprogramming a protease to cut a specific sequence requires a highly efficient and robust evolution system capable of handling a large amount of mutational diversity^{8,9,16,17}. To evolve protease specificity in the proteome, exploring a billion-level mutational space is usually necessary, which is challenging for the traditional fluorescent screening or yeast display approaches^{11,16,18}. This resulted in the evolution of proteases that have long been limited to enhancing activity and making minor alterations to the substrate sequences. Until recently, the phage-assisted continuous evolution (PACE) platform, which coupled protease activity to the lifecycle of bacteriophages, facilitated protease reprogramming that could achieve substantial changes in their substrate recognition sequences^{8,9}. However, mutations created using the PACE system are spontaneous and random

across the entire encoding gene of the protease, making it challenging to achieve the focused evolution on the protease's substrate recognition pocket¹⁹. Therefore, establishing a robust and straightforward selection method that can accomplish a saturated exploration of the mutational space in the substrate-binding pocket would be a major breakthrough for the development of protein-editing enzymes. One appealing approach is using a life–death selection system that directly links host growth to protease activity without any complex genetic pathway construction or multiple-step coupling; such a platform could effectively cover an extensive mutational space within a few rounds of selection with a minimalistic design to facilitate its adjustability, modularity and functionality.

To realize this vision, we report here an in vivo life–death selection system to reprogram the sequence recognition site of a protease. The selection system was optimized to be efficient, robust, stress tunable and capable of achieving millionfold enrichment of variants with the desired function in a single round of selection. We demonstrate that this approach can evolve the parent protease to reprogram its

recognition code and cleave different target proteins in living cells, including the membrane protein CD22, the mitochondrial protein ALAS1 and the nuclear protein EZHIP, highlighting its adaptability for broad applications.

Results

Development of selection system with tunable lethality based on N-terminal domain of gasdermin D (GD-N)

We aim to construct a two-component selection system in *Escherichia coli* to couple protease activity with host growth. In brief, we inserted the target cutting sequence into the loop region of GD-N, a toxic protein that induces growth stress (Fig. 1b)^{20–22}. Protease variants that recognize the target sequence destroy the toxic GD-N and the host bacteria harboring such variants become enriched in the liquid culture (Fig. 1b and Extended Data Fig. 1a).

The first objective was to identify a toxic protein that can provide tunable stress for *E. coli* growth. Inspired by the programmable cell death processes in mammalian cells, which is triggered by a cascade of caspase-mediated cleavage events, we prioritized building an artificial ‘programmable cell death’ system in *E. coli* by using the cell death trigger protein. We then focused on GD, a cell death trigger in pyroptosis. The full-length GD (GD-F) is activated by proteolytic cleavage releasing GD-N, which forms oligomers that create pores that serve as channels facilitating substance exchange across the cell membrane; this process is the key contributor to executing pyroptosis in mammalian cells^{21,23–25}. As this mechanism was well established in mammalian cells, we next tried to demonstrate whether the expression of GD-N in *E. coli* could induce cell death or cell growth stress (Fig. 1c).

To this end, we expressed the GD-N protein in *E. coli* and found that the growth was strongly suppressed (Fig. 1c and Extended Data Fig. 1b,c), as evidenced by the optical density at 600 nm (OD₆₀₀) of the bacteria expressing GD-N remaining below 0.1 after 12 h of culturing. By equipping this highly toxic GD-N protein, the evolution system could be used for the selection of a highly active protease from a low-activity parent variant. However, in the sequence specificity reprogramming of a protease, the initial activity of a protease mutant would be very low. Therefore, this highly toxic GD-N protein was not suitable for the initial evolution of a protease to recognize an alternative sequence. To establish a stress-tunable evolution system that could be applied for the selection of the weakly active protease against the alternative target sequences, we constructed a toxicity-attenuated GD-N protein by fusing a SUMO tag to the N terminus (SUMO–GD-N). The toxicity of SUMO–GD-N is moderate, which is suitable as the selection marker (Fig. 2b). We also added four substitutions (R137A, K145A, R151A and R153A) to SUMO–GD-N to construct a nontoxic variant or dead mutant^{20,26} of GD-N (SUMO–GD-N-4A) as the negative control (Fig. 1d,e and Extended Data Fig. 1d).

We next checked the mechanism of the toxicity from the GD-N protein. As expected, using a GFP-tagged construct (GD-N–GFP), we found that GD-N–GFP accumulated on the *E. coli* membrane (Extended Data Fig. 1e). We also observed that the expression of GD-N (Fig. 1f) and SUMO–GD-N (Fig. 1g and Extended Data Fig. 1f) triggered different degrees of membrane structural damage, consistent with their toxicity to bacterial growth^{21,23}. Additionally, we further determined that the selection stress from SUMO–GD-N could be tuned by altering its expression level and by changing the antibiotic concentration in the culture medium (Fig. 1h). This continuously tunable performance provides a wide selection window for the evolution of proteases with different activity levels, indicating that the SUMO–GD-N protein is a suitable inducer of selective stress to construct a protease evolution platform.

Validation of the GD-N-based protease reprogramming system

Next, to couple protease cleavage activity to host bacterial growth, we searched for a cleavage site on the SUMO–GD-N protein in which

to insert the target sequence for the evolved proteases that met the following requirements: (1) insertion of the sequence must not affect the toxicity of the GD-N protein; (2) the insertion sequence must be sufficiently exposed so that it can be efficiently cleaved by the protease; and (3) the cleaved GD-N protein fragment must lose its pore-forming ability to render it nontoxic to the host bacteria. Therefore, we analyzed the crystal structure of the GD-N protein and identified several potential positions for insertion of the target sequence^{21,22}.

Next, for proof of concept, we selected tobacco etch virus protease (TEVp) as our model protease to establish an evolutionary system for protease reprogramming. TEVp is characterized by a well-defined active site and a highly specific cleavage sequence, ENLYFQS, which is absent in both the *E. coli* and the human proteomes. This specificity confers a considerable advantage in minimizing off-target cleavage, making TEVp an ideal candidate for developing sequence-specific proteases through directed evolution^{27–29}. The ENLYFQS sequence was inserted into nine positions on the SUMO–GD-N protein either through direct insertion or through insertion–deletion to construct 15 variants (P1–P15) (Fig. 2a and Extended Data Fig. 2a,b).

We then evaluated the toxicity of the P1–P15 variants using SUMO–GD-N (wild type, WT) and the dead mutant as a control. We found that the P2, P4, P5, P9 and P12–P15 variants all failed to cause sufficient stress for *E. coli* growth, whereas the P1, P3, P6–P8, P10 and P11 variants caused remarkable selection stress to the host and this stress could be rescued by the expression of TEVp-WT (Fig. 2b and Extended Data Fig. 2c). To identify the most efficient SUMO–GD-N variant for protease evolution, we separately cotransformed the strains expressing P1, P3, P6–P8, P10 and P11 strains with TEVp-WT and mixed them in equal proportions into a single liquid culture. After the cultures were passaged three times at a 1:1,000 dilution, the proportion of P10-expressing cells was enriched to 53.1% of the culture (Fig. 2c,d and Extended Data Fig. 3a), prompting us to select the P10 variant for further application. Next, immunoblot analysis demonstrated that the P10 variant could be efficiently cleaved by TEVp-WT (Fig. 2e) and the cleavage of P10 effectively rescued the bacterial growth defect in both liquid and solid media (Fig. 2f,g). Together, these results confirm that host bacterial growth can be used as a readout of the specific protease activity of TEVp for target sequence cleavage.

Next, to explore the selection window of this evolutionary system, we compared the growth curves of *E. coli* coexpressing TEVp-WT and GD-N-P10 with those coexpressing the catalytically inactive TEVp mutant C151A (TEVp-C151A) and GD-N-P10 under various temperature conditions. We observed that the bacteria expressing TEVp-WT exhibited much faster growth rates than those expressing TEVp-C151A, providing a distinct selection window for evolving our proteases. Notably, this selection window demonstrated a pronounced temperature dependence; at lower temperatures, the growth curves of the *E. coli* were extended and, correspondingly, the selection window was also broadened (Fig. 3a–d). This characteristic allowed for fine-tuning of the evolutionary selection pressure through temperature adjustments, offering a convenient framework for the evolution of proteases with varying activities.

Subsequently, we evaluated the evolutionary efficiency of this platform for a fixed target sequence (ENLYFQS) and variable protease mutants. We conducted a model selection by mixing *E. coli* bacteria bearing TEVp-WT and catalytically inactive TEVp-C151A at a proportion ratio of 1:1–1:10⁶ and, after only one round of culture and selection, we sequenced the population and found that the bacteria bearing TEVp-WT was dominantly enriched (Fig. 3e–g and Extended Data Fig. 3b,c). Even when we started cultures with a 1:10⁵ proportion of TEVp-WT to TEVp-C151A, the final proportion of TEVp-WT reached 94:6, which represented millionfold enrichment during the selection process (Fig. 3g and Extended Data Fig. 3b,c). To evaluate the evolutionary efficiency of this platform for a fixed protease variant (TEVp-WT) and different target sequences, we performed a model selection by

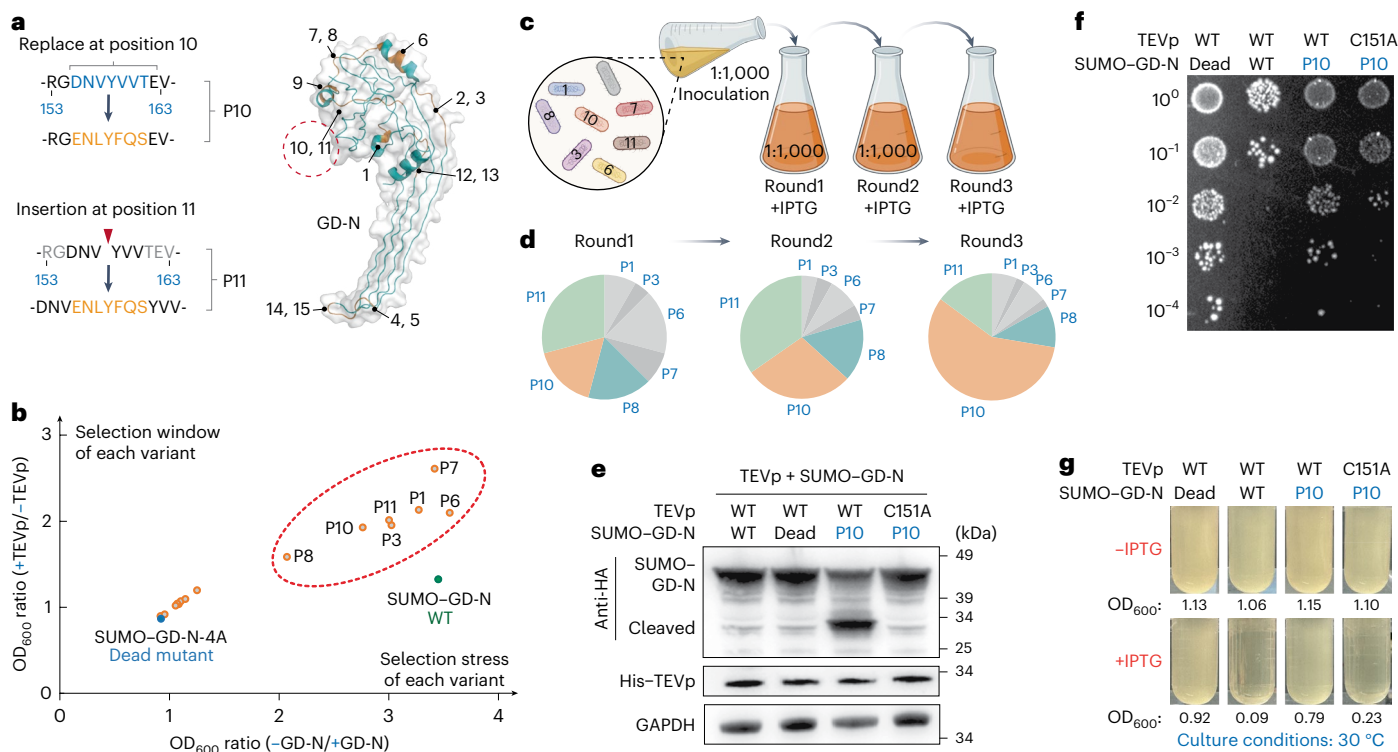


Fig. 2 | Establishment of the GD-N-based selection system for protease evolution. **a**, Structure of the GD-N protein (PDB 6VFE). The positions into which the TEVp-WT recognition sequence ENLYFQS was inserted or replaced are labeled on the structure. The corresponding variants of SUMO-GD-N were named P1–P15. Detailed information of each variant of P1–P15 can be found in Extended Data Fig. 2. **b**, Evaluation of the P1–P15 variants to identify suitable SUMO-GD-N variants for further applications. The x axis is the OD_{600} ratio between bacteria without the GD-N protein and bacteria expressing GD-N, representing the toxicity of each SUMO-GD-N variant. The y axis is the OD_{600} ratio between bacteria expressing TEVp-WT and bacteria without TEVp-WT expression, demonstrating the ability of TEVp-WT to cleave the target sequence

and restore bacterial growth. **c**, Scheme of the competition selection experiment, in which bacteria expressing P1, P3, P6, P7, P8, P10 and P11 were mixed in equal proportions. After several rounds of incubation, the proportions were analyzed by sequencing. **d**, The results of competitive selection showed that the proportion of the P10 strain gradually enriched over several passages. **e**, Immunoblotting analysis showing TEVp-WT expressed in bacteria efficiently cleaved the P10 variant of the SUMO-GD-N protein. **f**, A spot titer assay showed that P10 expression inhibited bacterial growth, which could be restored by the expression of TEVp-WT. **g**, Expression of TEVp-WT rescued the toxicity of GD-N in the LB culture system. Schematic diagrams were created with BioRender.com.

mixing *E. coli* expressing GD-N-P10 (ENLYFQS) and GD-N-WT at a 1:1–1:10⁶ ratio. Bacteria expressing GD-N-P10 were dominantly enriched 4.56 × 10⁵-fold after only one round of culture and selection (Fig. 3h–j and Extended Data Fig. 3d).

These results demonstrated that this life–death selection system could efficiently enrich the desired protease variant capable of cleaving the target sequence. However, the system cannot exclude those variants with both on-target and off-target cleavage capability. Therefore, to exclude evolved variants that are selected for on the basis of potent on-target activity but that have undesirable off-target activity as well, we designed a negative selection system by inserting any identified specific off-target sequences that do not want to be cut between the N-terminal domain and the C-terminal domain of GD-F²¹. If a TEVp variant has off-target activity that cleaves the GD-F insertion sequence, the released GD-N protein would trigger *E. coli* death, thereby preventing enrichment of this variant during the evolution processes. We, therefore, picked nine positions for the insertion of the WT-TEVp substrates (ENLYFQS) as a model off-target sequence (N1–N9) and used these variants for further screening. After coexpression of each GD-F variant and TEVp-WT, we identified three GD-F variants—N2, N3 and N7—that exhibited the expected toxicity, indicating that these three positions could be used to construct an efficient negative selection system (Extended Data Fig. 4). By using the negative selection system, an evolved protease variant that cut any defined off-target sequence will be avoided.

In summary, compared to other evolution systems that need complex genetic circuits or sophisticated infrastructure, these results

demonstrate that this method is an efficient, robust and stress-tunable platform capable of selecting the desired protease variant from millions to billions of possibilities after just a few rounds of selection in culture.

Evolution of TEVp to cleave the sequence from membrane-bound CD22

As a test of the utility of the protease evolution system, we selected a target sequence (EPVKVQH) from the membrane protein CD22 to reprogram the sequence specificity of TEVp. CD22 is a B cell-specific membrane protein and a druggable target for treating lymphoma and rheumatoid arthritis^{30–32}. Different drug modalities targeting CD22, including monoclonal antibodies, antibody–drug conjugates and chimeric antigen receptor T cell therapies, have been approved for therapeutic applications^{31,33,34}. We, therefore, envisioned that a sequence-specific protease capable of cleaving the extracellular domain of CD22 may have potential therapeutic benefits. Within the extracellular domain, we identified an exposed loop on domain 7 and selected EPVKVQH (647–653) as our target sequence to evolve TEVp (Fig. 4a)^{32,35}.

We generated a mutant CD22 where EPVKVQH (647–653) was replaced with the native TEVp target sequence (ENLYFQS) and we subsequently confirmed that this targeted loop region on CD22 is sufficiently exposed for protease cleavage (Fig. 4b). Because our target sequence and the TEVp-WT recognition sequence had different amino acids at five positions, we first evaluated the tolerance of TEVp-WT

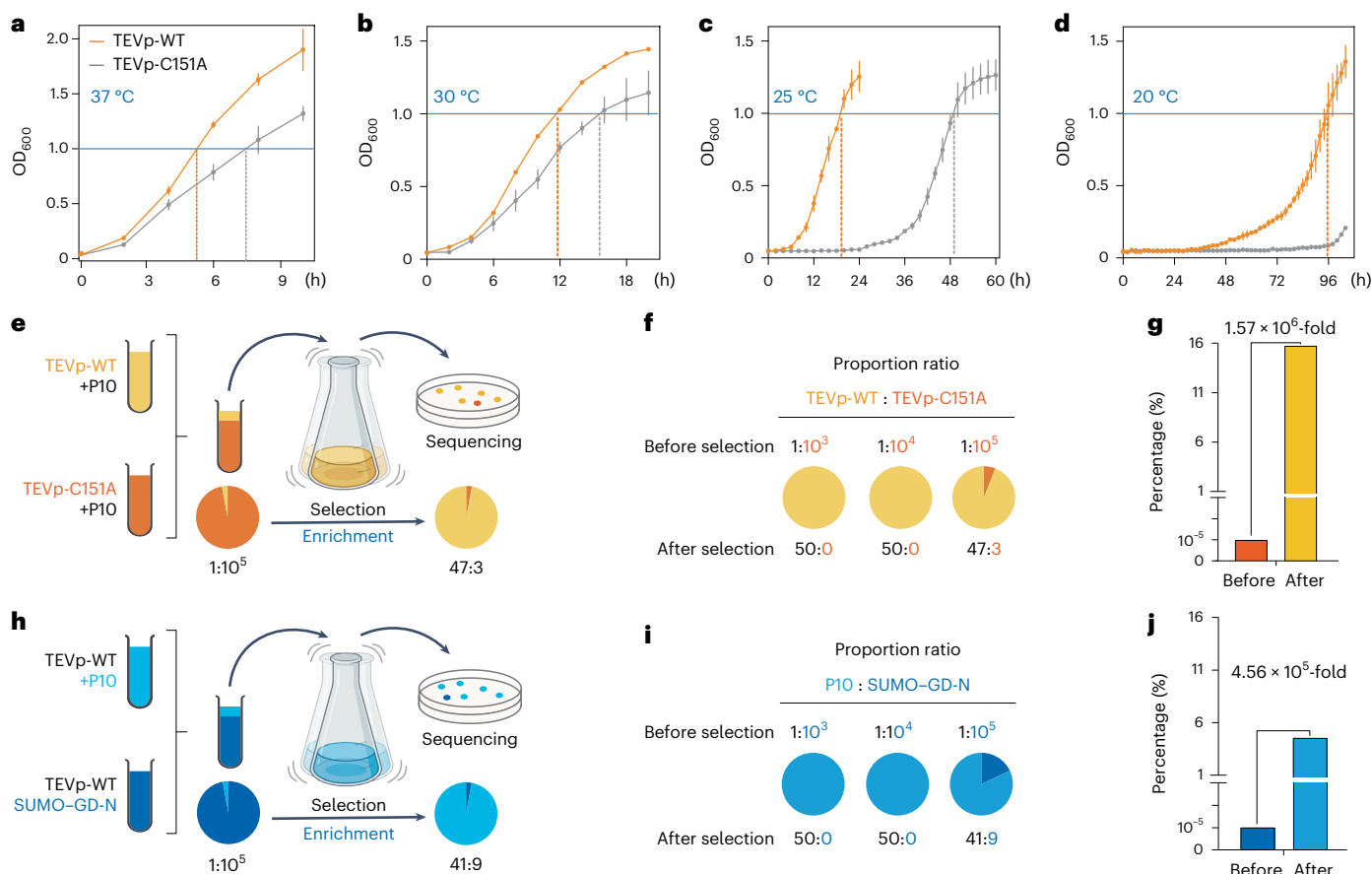


Fig. 3 | Million-fold enrichment of evolved proteases using the GD-N-based selection system. a–d, Growth of *E. coli* expressing TEVp-WT or the dead mutant could be strictly controlled by changing the temperature of the culture. **e–g,** Enrichment of TEVp-WT from a culture started with an extreme excess of the dead mutant. The enrichment reached 10⁶-fold in a single round of selection.

h–j, Enrichment of bacteria harboring TEVp-WT and GD-N-P10 in a culture started with an extreme excess of bacteria harboring a nonsubstrate variant of GD-N. The enrichment reached 10⁶-fold in a single round of selection. Schematic diagrams were created with BioRender.com.

to substitutions at each of these positions. We found that TEVp-WT showed promiscuous activity for single-amino-acid changes at the P1', P2, P3 and P5 sites but showed little activity when three or more amino acids were substituted simultaneously (Fig. 4c); therefore, we used a 'substrate-walking' evolutionary strategy and selected a series of substitution sites on the basis of the TEVp-WT–substrate complex structure (Fig. 4d,e and Extended Data Fig. 5a,b)^{8,28,29}. Before initiating the screening process, we evaluated the quality and transformation efficiency of all mutant libraries involved in this study. The results demonstrated that the saturation mutagenesis libraries exhibited excellent diversity with no notable amino acid bias. Additionally, the transformation efficiency was sufficient to meet the throughput requirements of each screening step (Extended Data Fig. 6). After two rounds of evolution, we obtained different TEVp mutants, among which TEV_{CD22}-2.1 demonstrated weak activity against the target sequence EPVKVQH (Fig. 4f and Extended Data Figs. 5 and 7). Subsequently, we constructed a new saturation mutagenesis library based on the TEV_{CD22}-2.1 variant for further selection and obtained a variant (TEV_{CD22}-3.1) that more efficiently cut the alternative sequence (Fig. 4f). Thus, TEVp could recognize the target sequences, even when five of seven amino acids of the WT substrate sequence were altered.

Evolution of TEVp to recognize and cleave mitochondrial ALAS1

We next selected ALAS1 as the target protein to evolve TEVp. ALAS1 is a mitochondrial protein that acts as the rate-determining enzyme of

heme biosynthesis (Fig. 5a, left)³⁶ and knockdown of ALAS1 in the liver mediated by small interfering RNA has been clinically applied to treat acute intermittent porphyria^{37,38}. In addition, ALAS1 was recently demonstrated as the key protein to limit microRNA accumulation and ALAS1 suppression in the liver generally enhances synthetic small interfering RNA silencing activity, indicating its potential as general adjuvant in RNA interference therapy³⁹. We hypothesized that our evolutionary protease selection platform could be used to reengineer TEVp precisely to cut mitochondrial ALAS1. Before we started evolution, we expressed an ALAS1 variant with the original TEVp-WT substrate sequence ENLYFQS replacing the native HNIYVQA sequence at positions 551–557 in HEK293T cells (Fig. 5a). This modified ALAS1 harboring ENLYFQS was then incubated with TEVp-WT and immunoblotting analysis revealed a 70-kDa band below the native ALAS1 band (Fig. 5a, right), confirming that the original sequence at the 551–557 position of ALAS1 was accessible and could be cleaved by TEVp-WT.

We then tested the ability of TEVp-WT to cleave a series of intermediate substrate sequences from ENLYFQS to HNIYVQA using a fusion protein composed of maltose-binding protein (MBP), substrate sequences and GFP (MBP–P6–N–P4–Y–P2–Q–P1'–GFP). TEVp-WT could not cleave the final MBP–HNIYVQA–GFP protein but showed weak activity toward three intermediate sequences (Fig. 5b), prompting us to design an evolutionary route for the final target sequence (Fig. 5c). We analyzed the substrate-binding pocket and constructed a saturation mutagenesis library for TEVp with substitutions in the substrate-binding pocket at four sites (Extended Data Fig. 8a). We

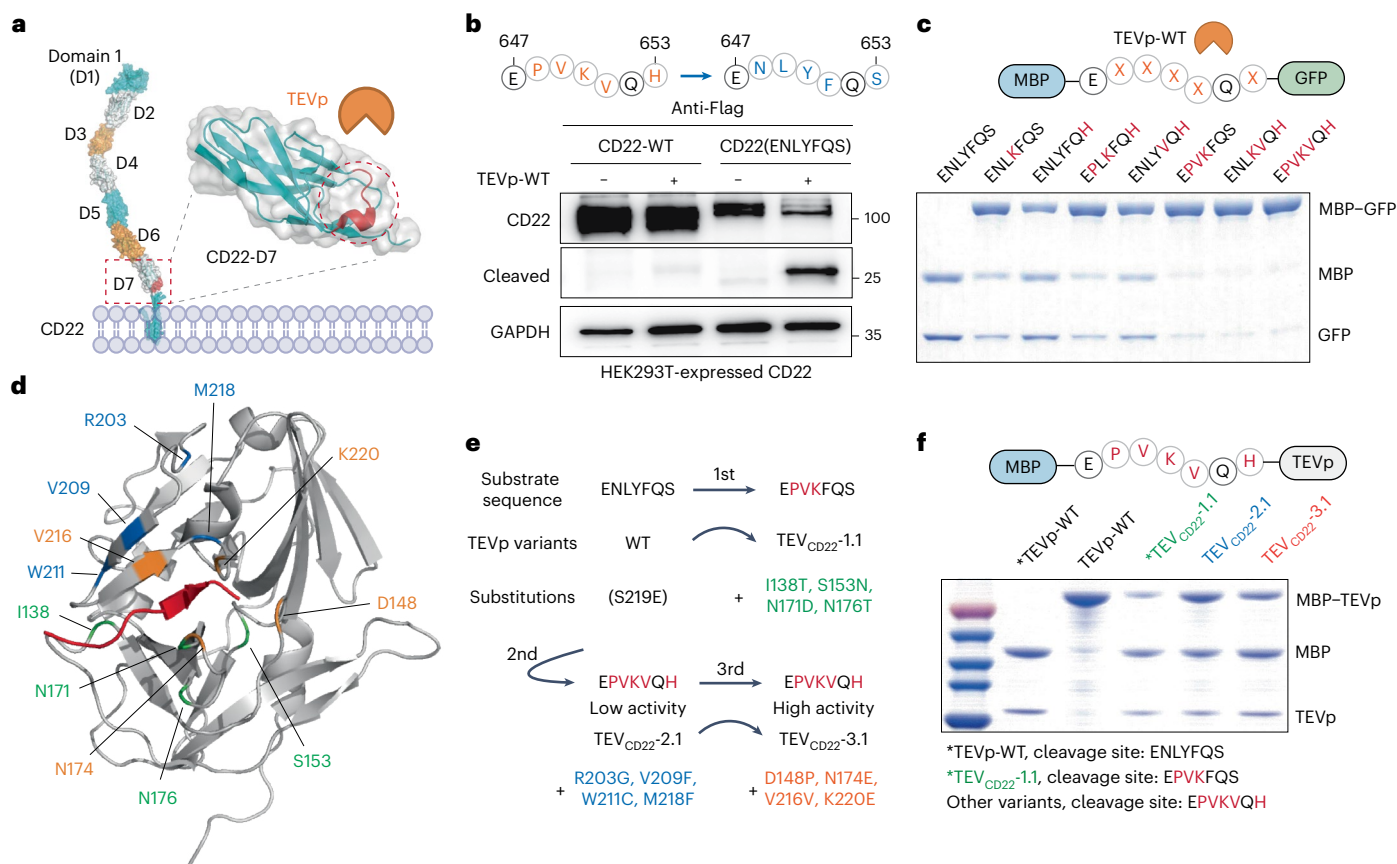


Fig. 4 | Evolution of TEVp-WT to cleave CD22. a, Structure and cellular location of CD22 (PDB 5VKJ and 7O52). **b**, Immunoblot verifying residues 647–653 of CD22 can be accessed by TEVp-WT by replacing the original CD22 sequence at this position (EPVKVQH) with TEVp-WT substrate sequence ENLYFQS. **c**, Immunoblot verifying the ability of TEVp-WT to cleave the target sequence (EPVKVQH) and corresponding intermediate sequences in the evolutionary pathway.

d, Structure and mutation sites of TEVp-WT substrate-binding pocket (PDB 1LVM). **e**, Designed evolutionary trajectories from the sequence recognized by TEVp-WT to the target sequence. **f**, Comparison of cutting efficiency of evolved TEVp variants for the target sequence using a self-cleavage recombinant protein (MBP-EPVKVQH/EPVKFQS/ENLYFQS-TEVp).

cotransformed this library with a selection system bearing the HNIYVQA sequence^{28,29} and, after the first round of selection, the surviving colonies were sequenced. Each site's enriched amino acids were shown as a pattern logo (Extended Data Fig. 8b,c and 9), revealing five potential hits that were subsequently evaluated individually for their ability to cut the HNIYVQA sequence. The TEV_{ALAS1}-1.2 variant exhibited approximately 20% cutting efficiency (Extended Data Fig. 8g) and we prioritized it for further optimization using a second-generation saturation mutagenesis library with an additional four amino acid substitution sites (Extended Data Fig. 8d). After this second-round selection, the enriched amino acids at each site were shown as a pattern logo (Extended Data Fig. 8e,f and 9), which uncovered four potential hits that were tested individually for their ability to cut the HNIYVQA sequence. The TEV_{ALAS1}-2.1 variant exhibited approximately 60% cutting efficiency over 3 h in the condition of 5 μ M substrates and 1% TEVp (Extended Data Fig. 8g,h). When we increased the concentration of substrates to 25 μ M, the cleavage efficiency could reach 90% (Extended Data Fig. 8i). Subsequently, we used the fluorogenic peptide substrates and the protein substrates to verify the specificity of the TEVp-WT and TEV_{ALAS1}-2.1; we found that TEV_{ALAS1}-2.1 showed much less activity than TEVp-WT for the protein substrates with ENLYFQS, whereas TEVp-WT showed no activity for the HNIYVQA sequence. These results verified that the specificity of TEVp was reprogrammed rather than evolved to be more promiscuous (Fig. 5d,e). In addition, we systematically evaluated the sequence tolerance at each position of the recognition motif individually for

both TEVp-WT and TEV_{ALAS1}-2.1 using our platform. The substrate recognition patterns differed substantially between the WT and evolved variants (Extended Data Fig. 8j,k). These results further confirmed that directed evolution altered the substrate specificity of TEVp rather than increasing its promiscuity.

Next, to verify the ability of TEV_{ALAS1}-2.1 to cut full-length ALAS1, we expressed WT ALAS1 protein in HEK293T cells. After incubating the cell lysate with TEV_{ALAS1}-2.1 for 3 hours, the expected 70-kDa band indicative of cleavage was observed through immunoblotting analysis (Fig. 5f). To verify the capability of TEV_{ALAS1}-2.1 to cleave in live cells, we fused a cell-penetrating artificial mitochondria-targeting peptide (CAMP) to the N terminus of TEV_{ALAS1}-2.1 (Fig. 5g) to deliver the protease into the mitochondria, where ALAS1 is located^{40,41}. We expressed ALAS1 into HEK293T and confirmed cleavage of intracellular ALAS1 by immunoblotting analysis (Fig. 5h). Lastly, to demonstrate whether this protease will generate off-target cleavage on the proteome, we performed a comprehensive quantitative proteomic analysis of cell lysates treated with TEV_{ALAS1}-2.1 and the cell treated with TEVp dead mutant was used as the control. The quantitative proteomic analysis of cell lysates treated with TEV_{ALAS1}-2.1 revealed no notable proteome-wide changes compared to treatment with the TEV dead mutant, suggesting minimal off-target effects of the evolved TEVp (Fig. 5i). Collectively, these results demonstrate the potential of TEV_{ALAS1}-2.1 to cleave ALAS1 protein in liver, which might be further developed as an alternative strategy for treating acute intermittent porphyria treatment or could be further used as a general adjuvant in RNA interference therapy.

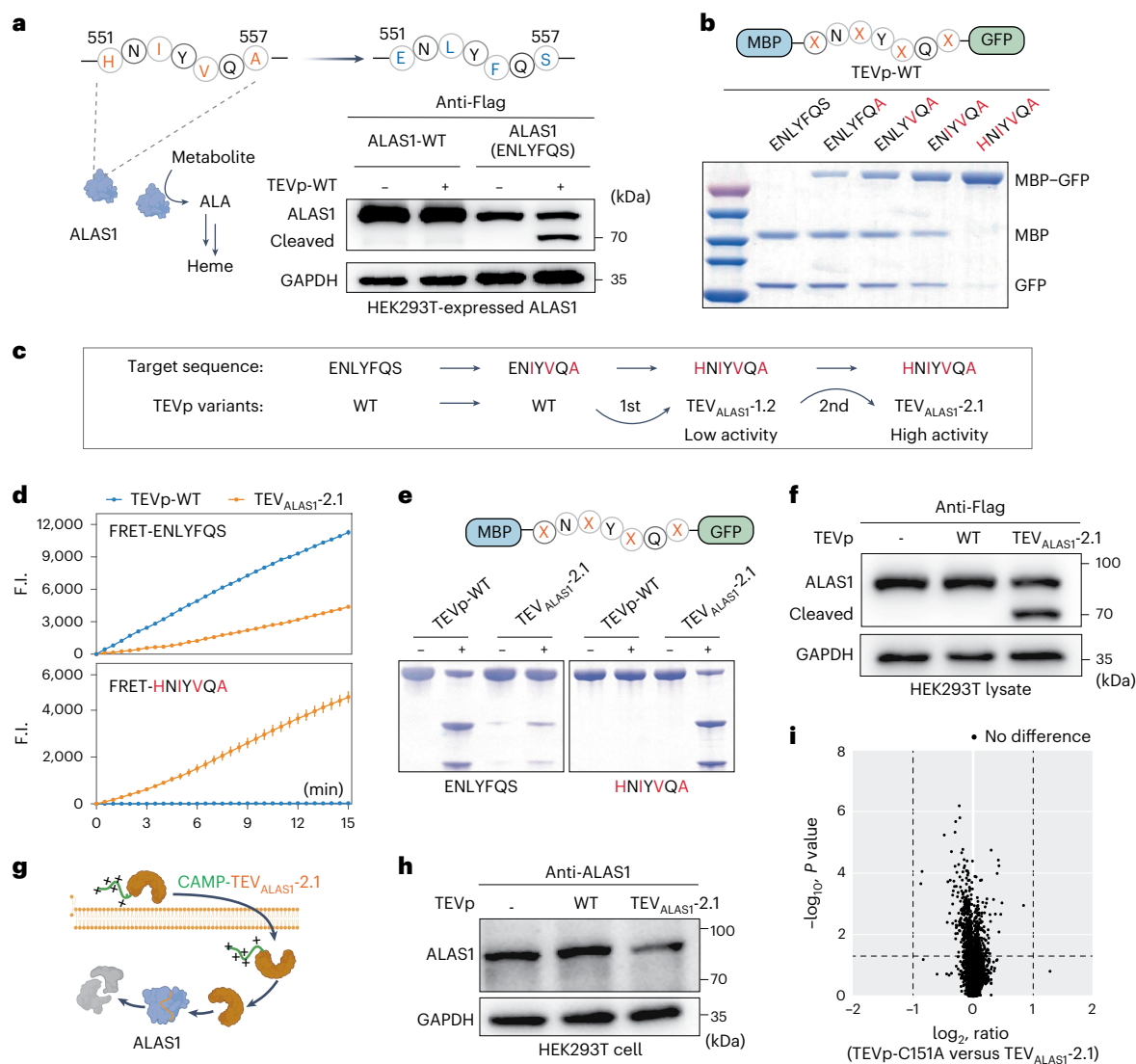


Fig. 5 | Evolution of TEVp to cleave ALAS1. **a**, Schematic of the location and function of ALAS1. Immunoblot verifying that residues 551–557 of ALAS1 can be accessed by TEVp-WT by replacing the original ALAS1 sequence at this position (HNIYVQA) with TEVp-WT substrate sequence ENLYFQS. **b**, Immunoblot verifying the ability of TEVp-WT to cleave the target sequence HNIYVQA and corresponding intermediate sequences in the evolutionary pathway. **c**, Schematic of the designed evolutionary path from the sequence recognized by TEVp-WT to the target sequence. **d, e**, Cross-comparison of cutting efficiency of evolved TEV_{ALAS1}-2.1 variant and TEVp-WT for both ENLYFQS and HNIYVQA substrate sequences using fluorogenic peptide or MPB-cleavage site-GFP protein. In **d**, the concentrations of the fluorogenic peptide and enzyme were 50 μ M and 1.2 μ M, respectively. In **e**, the concentrations of the protein substrate

and enzyme were 25 μ M and 5 μ M, respectively. **f**, Verification of the ability of the evolved TEVp variant (TEV_{ALAS1}-2.1) to cleave ALAS1 protein in mammalian cells. TEVp variants were added to the cell lysate and then immunoblot analysis was performed. **g**, Scheme of the delivery system for transporting the protease into the mitochondria of living cells, where endogenous ALAS1 resides. **h**, Immunoblotting analysis demonstrating cleavage of ALAS1 by TEV_{ALAS1}-2.1–CAMP in live cells. Because of the epitope of the primary antibody of ALAS1, the cleaved band cannot be observed; only the decrease of the original band can be observed. **i**, A comprehensive quantitative proteomic analysis of cell lysates treated with TEV_{ALAS1}-2.1 to demonstrate that the TEV_{ALAS1}-2.1 variants exhibit no notable off-target cleavage within the proteome. Schematic diagrams were created with BioRender.com.

Evolution of TEVp to recognize and cleave nuclear EZHIP

Next, we aimed to evolve TEVp-WT to recognize the ERLAFQS sequence at positions 377–383 of the enhancer of zeste homolog inhibitory protein (EZHIP). EZHIP is a nuclear EZH2-binding protein that inhibits the methylation activity of the RCC2 complex (Fig. 6a) and recent findings showed that the characteristic global oncogenic H3K27me3 mark in PFA ependymoma is caused by overexpression of EZHIP^{42–45}. Therefore, evolving a TEVp variant to target EZHIP could potentially reprogram the epigenetic homeostasis of neural cancer cells.

To evolve the protease to recognize the ERLAFQS sequence of EZHIP, we demonstrated that residues 377–383 of EZHIP were exposed and could be accessed by TEVp by substituting the TEVp-WT substrate sequence ENLYFQS at this position. EZHIP harboring the ENLYFQS

sequence was expressed in HEK293T cells and the cell lysate was incubated with TEVp-WT. Immunoblotting revealed a 50-kDa band, which was consistent with the expected molecular weight of cleaved EZHIP (Fig. 6a), indicating residues 377–383 of EZHIP were accessible to and could be cleaved by TEVp-WT.

We demonstrated that TEVp-WT fail to cleave the final MBP–ERLAFQS–GFP protein but showed weak activity toward two intermediate sequences (Extended Data Fig. 10b). As there is only a 1-aa difference between the intermediate sequence and the target sequence, we designed a saturation mutagenesis library at four sites and conducted the selection using the aforementioned procedure. We identified the enriched amino acids after one round of selection (Fig. 6b) and we also identified four TEVp variants that recognized the ERLAFQS

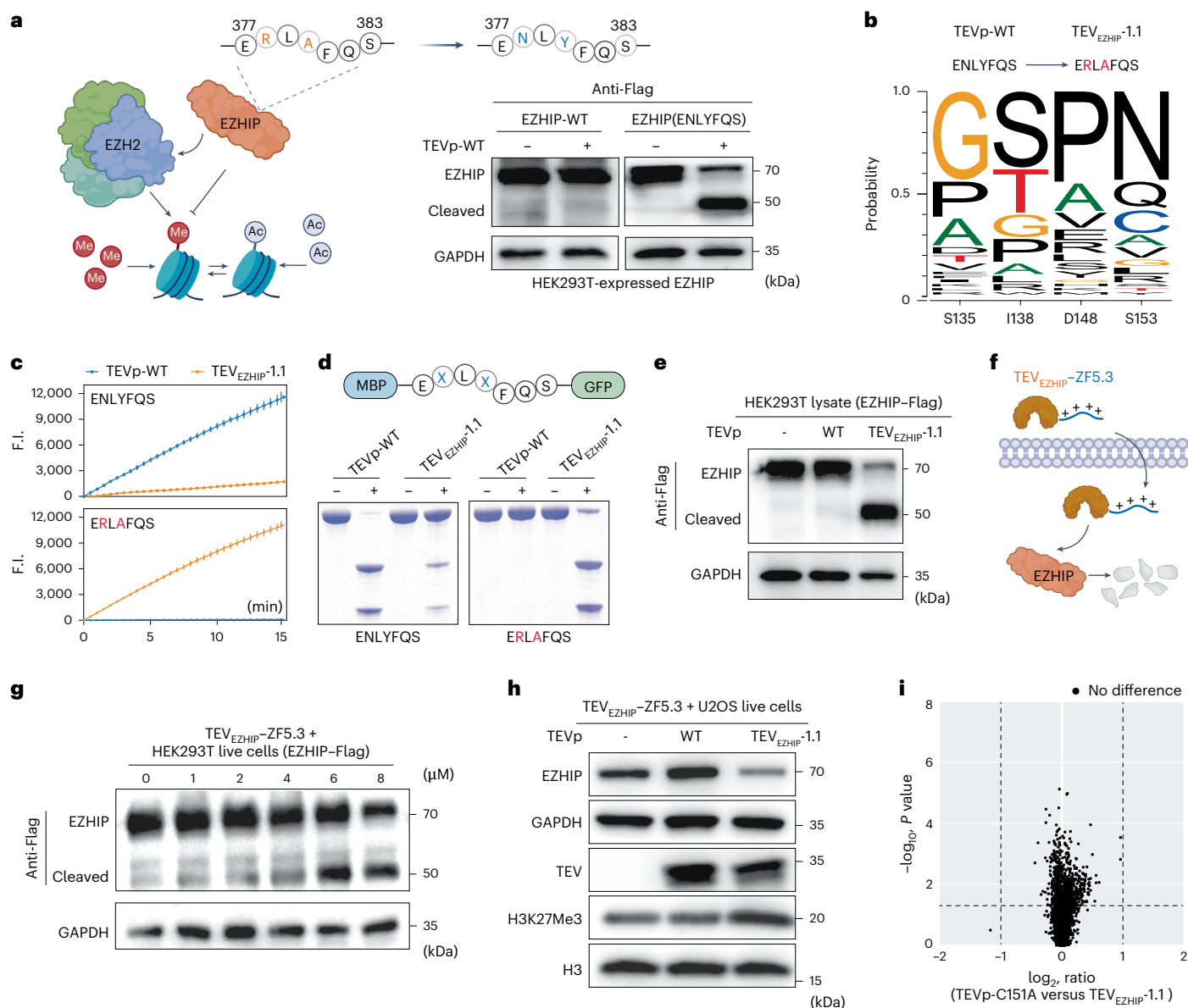


Fig. 6 | Evolution of TEVp for the cleavage of the intracellular EZHIP protein.

a, Schematic of EZHIP protein function. EZHIP is an EZH2-binding protein that inhibits the methylation activity of the RCC2 complex. Immunoblot verifying that residues 377–383 region of EZHIP can be accessed by TEVp-WT by replacing the original EZHIP sequence at this position (ERLAFQS) with the TEVp-WT substrate sequence ENLYFQS. **b**, Evolutionary pathway (top) and selection results (bottom) of TEVp-WT. Four sites were chosen for the construction of a saturated mutagenesis library. The sequence logo represents the frequency of each amino acid at the four sites of TEVp-WT after one round of selection. **c,d**, Cross-comparison of cutting efficiency of evolved TEV_{EZH2}-2.1 variant and TEVp-WT for both ENLYFQS and ERLAFQS substrate sequences using fluorogenic peptide or MPB-cleavage site-GFP protein. In **c**, the concentrations of the fluorogenic peptide and enzyme were 50 μ M and 1.2 μ M, respectively.

In **d**, the concentrations of the protein substrate and enzyme were 25 μ M and 5 μ M, respectively. **e**, Cleavage of the target protein EZHIP by TEV_{EZH2}-1.1 in cell lysates of HEK293T cells overexpressing EZHIP. **f**, Schematic illustration of the delivery of TEV_{EZH2}-1.1 into live cells by fusing with a ZF5.3 sequence. **g**, Immunoblot verifying that TEV_{EZH2}-1.1-ZF5.3 successfully cleaved the intracellular target protein EZHIP in live HEK293T cells overexpressing EZHIP. **h**, Immunoblot verifying that TEV_{EZH2}-1.1-ZF5.3 was delivered into U2OS cells to efficiently cleave the endogenously expressed EZHIP and the epigenetic H3K27 marker was further reconstructed. **i**, A comprehensive quantitative proteomic analysis of cell lysates treated with TEV_{EZH2}-1.1 to demonstrate that the TEV_{EZH2}-1.1 variants exhibit no notable off-target cleavage within the proteome. Schematic diagrams were created with BioRender.com.

sequence (Extended Data Fig. 10a–d). Among them, TEV_{EZH2}-1.1 had the highest cutting efficiency for the MBP-ERLAFQS-GFP substrate (Extended Data Fig. 10d,e) and was used for further study.

Subsequently, we used the fluorogenic peptide substrates and the protein substrates to verify the specificity of TEV_{EZH2}-1.1 variant. We found that TEV_{EZH2}-1.1 showed much less activity than TEVp-WT for the protein substrates with ENLYFQS, whereas TEVp-WT showed no activity for ERLAFQS sequence (Fig. 6c,d). In addition, we systematically

evaluated the sequence tolerance at each position of the recognition motif for both TEVp-WT and TEV_{ALAS1}-2.1 using our platform. The substrate recognition patterns differed substantially between the WT and evolved variants (Extended Data Fig. 8j and 10f). These results both verified that the specificity of TEVp was reprogrammed rather than evolved to be more promiscuous. To verify whether TEV_{EZH2}-1.1 could cleave the full-length EZHIP protein, we expressed WT EZHIP in HEK293T cells and incubated the cell lysate with TEV_{EZH2}-1.1 at 30 °C.

Immunoblotting analysis revealed that TEV_{EZH1P}-1.1 could cleave the EZHIP protein with more than 90% efficiency (Fig. 6e).

Next, to test the ability of TEV_{EZH1P}-1.1 to cleave endogenous EZHIP in living cells, we screened different delivery systems for shuttling TEV_{EZH1P}-1.1 into the cytoplasm. We tested two premixed cell-penetrating peptides and two genetically fused cell-penetrating tags; the results demonstrated that TEV_{EZH1P}-1.1 fused with the ZF5.3 tag exhibited better delivery efficiency (Fig. 6f,g and Extended Data Fig. 10g–j)^{46–50}. Although the WPC peptide also exhibited high delivery efficiency for TEV_{EZH1P}-1.1, the genetically fused approach could be a more convenient and broadly applicable method for other applications. We selected U2OS cells, which endogenously express EZHIP, as the model system to verify the cleavage efficiency of TEV_{EZH1P}-1.1 in living cells. Incubation of U2OS cells with TEV_{EZH1P}-1.1–ZF5.3 for 24 h resulted in the disappearance of full-length EZHIP, indicating complete cleavage of the protein, whereas no EZHIP cleavage was apparent after incubation with TEVp-WT. In addition, the cleavage of endogenous EZHIP by TEV_{EZH1P}-1.1–ZF5.3 reprogramed the H3K27 epigenetic marker, including methylation and acetylation (Fig. 6h). Lastly, to demonstrate whether TEV_{EZH1P}-1.1–ZF5.3 will generate off-target cleavage on the proteome, we performed a comprehensive quantitative proteomic analysis of cell lysates treated with TEV_{EZH1P}-1.1. As expected, the quantitative proteomic study revealed that TEV_{EZH1P}-1.1 showed no notable proteome-wide changes compared to treatment with the TEV dead mutant, suggesting minimal off-target effects of the evolved TEVp (Fig. 6i). Altogether, these findings verify that TEV_{EZH1P}-1.1 could cleave endogenous, intracellular EZHIP, again demonstrating the broad applicability of the selection platform to evolve highly specific proteases for potential clinical applications.

Discussion

In this study, we developed an evolution platform to reprogram TEVp-WT specificity to expand the breadth of substrates that could be efficiently cleaved. We first evolved TEVp-WT to target the B cell-specific membrane receptor CD22, which is a critical ligand for immune response and B cell signaling transduction. The target sequence on CD22 harbored a 5-aa difference compared to the native TEVp-WT recognition sequence ENLYFQS, indicating that the recognition sequence of proteases can be markedly changed by this evolution system to recognize virtually any target sequence of interest within the proteome. We next demonstrated the evolution of TEVp to cleave the mitochondrial protein ALAS1, which is an essential protein for heme biosynthesis, and the nuclear protein EZHIP, which regulates histone methylation. These two cases confirmed protein targets within different intracellular membranes could also be cleaved by an evolved TEVp. Notably, we achieved targeted cleavage of these endogenous intracellular proteins in pathologically relevant cell lines, demonstrating the potential of this platform for in vivo proteome editing.

We demonstrated that the target sequences on these proteins could differ by up to 5 aa from the original TEVp-WT substrate sequence, indicating that our evolution system can broadly reprogram the specificity of proteases to enable precise manipulation of protein functions and interactions. Combining a substrate-walking strategy and our protease evolution platform could be extended to other proteases and targets, broadening the technology's applications^{8,9}. We also provide proof of concept that multiple protein delivery systems could facilitate targeted intracellular cleavage of endogenous proteins in live cells^{46,50}.

Analogous to the natural evolution process of antibodies, which are selected in vivo from diversities ranging between 10¹⁰ and 10¹² to bind specific protein epitopes, we propose that directed evolution is a viable method for developing epitope-specific proteases for precise protein editing⁵¹. Just as B cells undergo extensive selection to produce highly specific antibodies, our evolution platform can achieve millionfold enrichment in a single selection round. With multiple rounds of library building and screening, our method can cover diversities

exceeding 1 billion variants of proteases, theoretically enabling the evolution of proteases specific to virtually any target sequences. Considering the success of engineering antibodies for therapeutic applications, we envision that the field of evolved protease recognition holds vast potential for growth, with promising avenues for developing therapeutic, diagnostic tools and beyond.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41589-025-02063-3>.

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Methods

Materials

All enzymes used in PCR progresses were ordered from New England Biolabs (NEB) and Vazyme (P501-d1). Primers were ordered from Sangon Biotech. Antibodies were purchased from Abcam, Proteintech and Cell Signaling Technology. Lipofectamine 3000 (transfection reagent) was purchased from Invitrogen. Expression vectors pET28(b) and pET11(a) with modifications were used and cloning was performed using Gibson assembly. *E. coli* cells were grown using Luria–Bertani (LB) medium with 100 $\mu\text{g ml}^{-1}$ ampicillin or 50 $\mu\text{g ml}^{-1}$ kanamycin. DH5 α chemically competent cells, DH10B electroporation-competent cells and BL21(DE3) T7 express competent *E. coli* cells were purchased from the NEB.

Spot titer experiments

Picked a single colony of *E. coli* into 3 ml of LB medium and incubated overnight at 37 °C with shaking at 220 rpm. The culture was placed on ice and *E. coli* cells were collected by centrifugation (4 °C, 3,000 rpm, 10 min). The pellets were suspended in PBS whilst adjusting OD₆₀₀ to approximately 1.0. Serial tenfold dilutions were made and, for every sample, 2 μl of each dilution was spotted on LB agar plates supplemented with antibiotics and inducers. The spots were allowed to dry. Plates were incubated in 37 °C for 12–16 h.

Membrane staining and imaging

The BL21(DE3) containing recombinant plasmid pET28a-GD was induced by 2% arabinose and 0.5 mM IPTG at 30 °C for 16 h. The induced and control uninduced *E. coli* cells were suspended in 1 ml of 0.1 M sodium bicarbonate (OD₆₀₀ = 1.0, pH 9.0) and stained with 500 μM FITC, respectively, for 1 h in the dark at room temperature. Then, the *E. coli* cells were centrifuged at 4,500 rpm for 5 min, washed three times with PBS and then resuspended with 60% glycerol. Confocal imaging was performed on an Axiovert 800 using 495 nm and $\times 63$ oil immersion objective.

Identification of suitable SUMO–GD-N variants for positive selection

We generated a modified pET11(a) vector that deleted the LacI promoter and LacI regions. The mutagenesis plasmid encoding 6 $\times$ His-tagged TEVp (containing S219V or S219E substitution) was assembled using the Gibson assembly protocol, named pTEV-S219V or pTEV-S219E. The expression vectors pET28(b) encoding SUMO–GD-N and a dead mutant variant with four substitutions (R137A, K145A, R151A and R153A) were constructed. The positions (P1–P15) inserted or replaced with the TEVp-WT recognition sequence ENLYFQS according to the structure of the GD-N protein (Protein Data Bank (PDB) 6VFE) and 15 selection plasmids (P1–P15) were generated. These selection plasmids were transformed into BL21(DE3) chemically competent cell with or without pTEV-S219V by heat shock. A single colony of each type of strain was picked into 3 ml of LB medium and incubated overnight at 37 °C with shaking at 220 rpm. The culture was inoculated into fresh LB medium contained antibiotic in a 1:1,000 ratio and cultivated at 37 °C with shaking. After 2 h, TEVp and SUMO–GD-N expression were induced with 0.5 mM IPTG. The group without IPTG was as control. The OD₆₀₀ values of each culture were determined every 2 h.

Negative selection

The expression vector pET28(b) encoding SUMO–GD was constructed. In this vector, positions N1–N9 were either inserted with or replaced by the TEVp-WT recognition sequence ENLYFQS on the basis of the structure of the GD protein (AF-P57764-F1). A total of nine negative selection plasmids (N1–N9) were generated. These plasmids were transformed into BL21(DE3) chemically competent cells with or without the pTEV-S219V plasmid by heat shock. Single colonies of each strain were picked and cultured overnight at 37 °C in 3 ml of LB medium with

shaking at 220 rpm. The overnight culture was diluted 1:1,000 into fresh LB medium containing the appropriate antibiotic and grown at 37 °C with shaking. After 2 h, protein expression of TEVp and SUMO–GD was induced with 0.5 mM IPTG. Cultures without IPTG served as controls. OD₆₀₀ values of each culture were monitored every 2 h to assess growth dynamics and protein expression.

Competition selection experiment

The selection plasmids (P1, P3, P6, P7, P8, P10 and P11) were transformed into BL21(DE3) chemically competent cells with pTEV-S219V using heat shock and placed on LB agar plates (100 $\mu\text{g ml}^{-1}$ ampicillin and 50 $\mu\text{g ml}^{-1}$). A single colony of each strain was placed into 3 ml of LB medium and incubated overnight at 37 °C with shaking at 220 rpm. The culture was collected by centrifugation (4 °C, 3,000 rpm, 10 min). Pelleted cells were suspended in sterile PBS whilst adjusting the OD₆₀₀ to approximately 1.0. All strains were mixed in equal proportion and incubated into fresh LB medium containing antibiotic in a 1:1,000 ratio. After 2 h, 0.5 mM IPTG was added. After three rounds of incubation and induction, the proportion of each strain was analyzed by Sanger sequencing.

Selection window optimization and model selection progress

To explore the selection window of this evolutionary system, selection plasmid P10 was transformed into BL21(DE3) chemically competent cells with pTEV-S219V or pTEV-S219V(C151A). pTEV-S219V encoded WT TEVp that could cleave GD-N-P10. The host cell TEV-WT + P10 would survive after induction and cleavage. pTEV-S219V(C151A) encoded a catalytically inactive TEVp mutant harboring a C151A substitution. The growth of host cell TEV-C151A + P10 would be suppressed after induction. All strains were incubated into fresh LB medium containing antibiotic and inducer in a 1:500 ratio. These cultures were incubated at 20, 25, 30 and 37 °C, respectively. The OD₆₀₀ values of each culture were detected every 2 h. To assess the evolutionary efficiency of this platform, a model selection by mixing BL21(DE3) cells bearing different combinations of plasmids (Supplementary Table 1) at a proportion ratio ranging from 1:10³ to 1:10⁶ for further culture. All mixed bacterial cultures were incubated into fresh LB medium containing antibiotic and inducer in a 1:200 ratio. The cultures were incubated at 25 °C with shaking at 220 rpm until the OD₆₀₀ reached approximately 0.3 to 0.5. The bacterial culture was diluted with PBS and plated on an LB agar plate (100 $\mu\text{g ml}^{-1}$ ampicillin and 50 $\mu\text{g ml}^{-1}$ kanamycin). A total of 50 single colonies were picked and Sanger sequencing was performed. The genotypes of TEVp and GD-N were counted.

Expression and purification of TEVp

TEVp was purified as previously described with minor modifications¹⁶. In short, we constructed expression vector pET28(b) encoding 6 $\times$ His-tagged TEVp fused through a TEV cleavage site to an MBP tag. BL21(DE3) chemically competent cells were transformed with the above constructs by heat shock. *E. coli* culture was grown in LB medium (400 ml) with kanamycin (50 $\mu\text{g L}^{-1}$) at 37 °C until the OD₆₀₀ reached approximately 0.8, followed by induction with 0.5 mM IPTG for 16 h at 20 °C. Then culture was pelleted and resuspended in 20 ml of binding buffer (30 mM HEPES pH 8.0, 500 mM NaCl, 30 mM imidazole and 10% glycerol). The resuspension was lysed by ultrasonication and then clarified by centrifugation at 18,000g for 30 min. The supernatant was collected and incubated with 5 ml of Ni-NTA agarose bead slurry (Cytiva) for 15 min at 4 °C. The slurry was transferred into a gravity column and washed with washing buffer (30 mM HEPES pH 8.0, 500 mM NaCl, 80 mM imidazole and 10% glycerol). The protein was eluted with elution buffer (30 mM HEPES pH 8.0, 500 mM NaCl, 300 mM imidazole and 10% glycerol). The purified protease was analyzed by SDS–PAGE and Coomassie blue staining. The eluted samples were dialyzed overnight in storage buffer (50 mM Tris-HCl pH 8.0, 100 mM NaCl and 2 mM EDTA) at 4 °C using Amicon Ultra-4 centrifugal filters (Ultracel,

nominal molecular weight limit: 10,000; Merck). Proteases used in mammalian cell culture were further subjected to endotoxin removal resin (Thermo Fisher Scientific). The concentration of each protein sample was determined using a Pierce BCA protein assay kit (Thermo Fisher Scientific) and DTT was added at a final concentration of 2 mM before freezing in liquid nitrogen for storage at -80°C in storage buffer (50 mM Tris-HCl pH 8.0, 10% glycerol and 2 mM DTT).

Cloning, expression and purification of MBP-TEVcs-GFP

The protease substrate MBP-TEVcs-eGFP (where TEVcs is either ENLYFQS or corresponding intermediate sequences in the evolutionary pathway) in pET28(b)+ vector was expressed in BL21(DE3) chemically competent cells¹¹. Cells were grown, induced and lysed as described above for TEVp. Ni-NTA purification was carried out in the same way. Fractions with bright-green protein were collected and transferred to an Amicon Ultra-15 centrifugal filter and exchanged three times into ice-cold 50 mM Tris-HCl buffer (pH 8.0) containing 10% glycerol and 1 mM EDTA. The concentration of each protein sample was determined using a Pierce BCA protein assay kit (Thermo Fisher Scientific) and DTT was added at a final concentration of 2 mM before freezing in liquid nitrogen for storage at -80°C in storage buffer.

Protease characterization

For the detection of TEV mutant activity, 28.8 μg of MBP-TEVcs-GFP substrate and 3.2 μg of TEVp were mixed in storage buffer (total volume: 80 μl) and incubated for 2 h at 30°C . The reaction was analyzed by SDS-PAGE and visualized with Coomassie blue stain. For the kinetics assay, the hydrolysis system was the same as above, except that the incubation was terminated at various time points.

Fluorogenic peptide substrate for FRET assay

A FRET-based fluorogenic peptide substrate was specifically designed to evaluate the *in vitro* proteolytic activity of TEVp. The custom-synthesized peptides, E(EDANS)-ENLYFQS-K(Dabcyl)-NH₂, E(EDANS)-HNIYVQA-K(Dabcyl)-NH₂ and E(EDANS)-ERLAFQS-K(Dabcyl)-NH₂ (purchased from Sangon Biotech) were used as the substrates for the assay. These peptides comprise 7 aa spanning the P6 to P1' positions relative to the cleavage site of TEVp. Following the addition of TEVp, fluorescence spectra were recorded immediately using a fluorescence spectrophotometer with excitation at 336 nm and emission at 490 nm. Measurements were taken at 30-s intervals over a period of 1 h.

Site-saturation mutagenesis

Site-saturation mutagenesis experiments were performed with primers bearing degenerate codons (NNK) using a modified QuikChange protocol³². The degenerate primers targeting the site of interest (forward and reverse) were used to yield one fragment by PCR amplification (Supplementary Table 2), which was further cyclized using T4 polynucleotide kinase and T4 DNA ligase. The PCR conditions were as follows (final concentrations): 1 $\times$ Phanta Flash master mix, 10 μM of forward primer, 10 μM of reverse primer and 0.1 ng μl^{-1} plasmid. Thermocycling conditions are outlined below (Supplementary Table 3). Upon completion of PCR, the remaining template DNA was digested with DpnI (1 μl). Gel purification was performed using gel electrophoresis (1% agarose gel containing a SYBR gold nucleic acid gel stain) and DNA was visualized on a blue transilluminator. DNA was isolated using the Thermo Scientific GeneJET DNA gel recovery kit. The purified DNA products were phosphorylated using T4 polynucleotide kinase. The phosphorylation conditions were as follows (final concentrations): 1 $\times$ T4 polynucleotide kinase reaction buffer, 10 U per μl T4 polynucleotide kinase, 1 mM adenosine 5'-triphosphate and 30 ng μl^{-1} DNA products. The thermal conditions were as follows: 37°C for 30 min for phosphorylation and 65°C for 20 min for thermoinactivation. All phosphorylated products were cyclized using T4 DNA at 4°C for 16 h. All products were transformed into DH10B electroporation-competent cells for plasmid amplification.

The complexity of mutant libraries

All four site-saturation mutagenesis libraries underwent Sanger sequencing to evaluate the quality and accuracy of the library construction. Sequencing chromatograms obtained from Sanger sequencing were analyzed to confirm the presence and uniformity of mutations. Subsequently, the DNA of each mutant library was transformed into DH10B competent cells. A total of 50 single colonies from each library were picked, Sanger-sequenced and analyzed for amino acid diversity. The WebLogo representing amino acid diversity was generated using an online tool (<https://weblogo.threeplusone.com/>). For further analysis, the 4M1 library was selected for next-generation sequencing (NGS). The plasmid library was subjected to PCR amplification targeting regions containing the mutation sites. Amplicon library preparation, sequencing and initial data processing were conducted by GENEWIZ. The PCR products were purified and underwent adaptor ligation for NGS library construction. Sequencing was performed on the Illumina NovaSeq platform (2×150 bp). The raw sequencing data were processed using GENEWIZ's standard bioinformatics pipeline, resulting in a sequence diversity analysis file. Further processing, analysis and visualization of the data were performed using R programming to elucidate the diversity and distribution of mutations within the library.

Production of electroporation-competent *E. coli* cells with selection plasmid

BL21(DE3) chemically competent cells were transformed with several selection plasmids; the cells carrying selection plasmids were named *E. coli* (P10*), where P10* indicates P10 modified with different target sequences. The sequence information is outlined in Supplementary Table 4. A single colony of *E. coli* (P10*) was placed in 3 ml of LB medium and incubated overnight at 37°C with shaking at 220 rpm. Then, 1 ml of cell culture was inoculated into 100 ml of LB medium with 50 $\mu\text{g ml}^{-1}$ kanamycin. The culture was incubated at 37°C with shaking at 220 rpm until the OD₆₀₀ reached approximately 0.5. Then, the culture was chilled on ice for 30 min. The culture was transferred into prechilled centrifuge bottles and centrifuged at 4°C for 15 min at 2,000g. The supernatant was decanted and the pellet was carefully resuspended in 100 ml of ice-cold 10% glycerol. The above step was repeated twice. The cleaned cell pellet was resuspended in 400 μl of ice-cold 10% glycerol, flash-frozen and stored at -80°C .

Selection process of TEVp mutant libraries

The selection processes were performed to evolve TEVp variants to cleave different target sequences. The site-saturation mutagenesis plasmid library was transformed into *E. coli* (P10*) according to the target sequence and selection step. About 500 ng of plasmids were mixed with 200 μl of *E. coli* (P10*) and transferred into ice-cold 0.2-cm electroporation cuvettes. After electric shock with 2.4 kV, the cells were diluted with 1 ml of SOC medium, cultured at 37°C for 1 h and then plated on LB agar plates (100 $\mu\text{g ml}^{-1}$ ampicillin and 50 $\mu\text{g ml}^{-1}$ kanamycin). After overnight cultivation at 37°C , the cells were eluted from the agar plate using LB medium. Then, 200 μl of cell culture was inoculated into 100 ml of LB medium (100 $\mu\text{g ml}^{-1}$ ampicillin and 50 $\mu\text{g ml}^{-1}$ kanamycin) with 0.5 mM IPTG. The culture was incubated at 25°C with shaking at 220 rpm until the OD₆₀₀ reached approximately 0.3 to 0.5. The bacterial culture was diluted with PBS and plated on LB agar plate (100 $\mu\text{g ml}^{-1}$ ampicillin and 50 $\mu\text{g ml}^{-1}$ kanamycin). A total of 50 single colonies were picked and Sanger sequencing was performed. The sequence information of the evolved TEVp variants was recorded.

Profiling protease sequence specificity in *E. coli*

To profile the substrate specificity of TEVp variants, we used a modified GD-directed evolution platform to examine TEV-WT, TEV_{ALAS1}-2.1 and TEV_{EZH1P}-1.1. In this system, the GD-F insertion sequences on the P10 plasmid were altered to introduce 20 distinct amino acid substitutions at

the P6 position of the TEVp cleavage site, individually. The 20 resulting plasmids were mixed in equal proportions to create the P6mix library, which was subsequently cotransformed with TEVp into competent cells for selection (the same approach was applied for positions P5 to P1). A total of 50 single colonies from each library were isolated, Sanger-sequenced and analyzed to determine substrate specificity. The WebLogo of amino acid sequence specificity was generated using an online tool (<https://weblogo.threeplusone.com/>).

Recombinant CD22, EZHIP and ALAS1 cleavage in vitro

The hCD22 cDNA clone was obtained from Sino Biological and hCD22–GFP was cloned into the pCMV3-C-Flag vector. The hEZHIP and hALAS1 cDNA clones were also obtained from Sino Biological and cloned into pCMV3-N-Flag vector separately. hCD22 (ENLYFQG)–GFP, hEZHIP (ENLYFQG) and hALAS1 (ENLYFQG) were generated by site-directed mutagenesis through a PCR procedure and separately cloned into pCMV3-C-Flag and pCMV3-N-Flag vectors. HEK293T and HepG2 cell lines were grown in DMEM with 10% FBS. The U2OS cell line was grown in McCoy's 5A medium with 10% FBS. All cell lines were cultured at 37 °C incubated with 5% CO₂. HEK293T cells were plated on a six-well plate and transfection was performed with Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions using 2.5 µg of each plasmid required for the specific experiment. HEK293T cells, which were transfected and incubated for 36 h, and HepG2 cells were scraped by cell scraper and transfer to a new tube. The cells collected in two wells of a six-well plate were added to 300 µl of cell lysis buffer (Beyotime), disrupted by an ultrasonic cell disruptor and then centrifuged at 12,000 rpm for 2 min at 4 °C. For the TEVp and variant cleavage experiment in vitro, 300 µl of reaction solution (100 µl of supernatant, 5.56 µM TEVp or variants, 50 mM Tris-HCl pH 8.0, 10% glycerol, 2 mM DTT and 1 mM PMSF) was incubated at 30 °C for 4 h.

Construction of TEVp–ZF5.3 plasmids and purification of TEVp–ZF5.3 fusion protein

The sequence of ZF5.3 was cloned into the C terminus of the sequence of TEV with linker GGGs to prepare the plasmid pET28a-TEVp-ZF5.3. Plasmids were transformed into *E. coli* BL21(DE3) for further expression. A single clone was inoculated into 5 ml of LB liquid medium (50 µg ml^{−1} kanamycin) and grown overnight. The cell cultures were transferred into 400 ml of LB liquid medium (50 µg ml^{−1} kanamycin) at a ratio of 1:100. All the flasks were shaken at 220 rpm at 37 °C. When the OD₆₀₀ reached 0.6–0.8, gene expressions were induced using 0.5 mM IPTG at 20 °C for 24 h. Subsequently, *E. coli* cells were harvested by centrifugation (4 °C, 6,000 rpm, 15 min), resuspended in 25 ml of buffer A (30 mM HEPES, 500 mM NaCl, 10% glycerol, 30 mM imidazole and 2 mM 2-mercaptoethanol) and subsequently followed by disruption with ultrasonic cell disruptor. The disrupted cells were centrifuged (4 °C, 6,000 rpm, 20 min) and the supernatants were further filtered through Millipore express membrane filters (0.45 µm; Sterile Millex filter unit; Merck). The filtered cell lysates were purified by His-Spin protein mini-prep columns. Purified formolase variant protein was subsequently concentrated and purification in a centrifugal filter unit (Amicon Ultra-15, 10 kDa; Merck kGaA, Darmstadt, Germany). The concentration of protein was determined using a protein assay kit (Micro BCA protein assay kit, Sangon Biotech) with BSA as the standard. The homogeneity of the purified sample was controlled by SDS–PAGE using standard molecular biology techniques.

Delivery of TEVp

HEK293T cells were transfected with plasmids expressing hEZHIP for 24 h and U2OS cells were seeded for 5 h in a 12-well plate. Different concentrations of TEVp-WT and variants with cell-penetrating peptide ZF5.3 were added to the medium of HEK293T or U2OS cells, which were then incubated for 12 h or 24 h separately and collected for western blot analysis.

Western blot analysis

Samples were electrophoresed on 10% Tris–glycine gels and then transferred to a PVDF membrane. Membranes were incubated at room temperature for 1 h in TBST with 5% nonfat milk and then washed three times. Primary antibody was added to TBST buffer in 1:1,000 dilution and the membranes were incubated on a rocker at 4 °C overnight (Supplementary Table 5). After three washes with TBST, the membranes were incubated 1 h at room temperature in TBST containing a 1:1,000 dilution secondary antibody. After three more washes with TBST, the membranes were scanned using a Tecan imaging system.

Proteomics analysis

To evaluate potential off-target proteolysis by TEV_{ALAS1}-2.1 or TEV_{EZHIP}-1 variants, proteomics analysis was performed according to previous methods with modifications^{53–55}. After cell lysate and protein extraction, the proteomics study was performed to evaluate whether the TEV_{ALAS1}-2.1 or TEV_{EZHIP}-1 variants resulted in notable off-target cleavage in the proteome. The HEK293T cell lysate was treated with TEVp-WT, TEV_{ALAS1}-2.1 or TEV_{EZHIP}-1.1 for 1 h and then subjected to trypsinization, tandem mass tag (TMT) labeling and quantitative proteomic analysis. The sample treated with TEVp dead mutant was used as the control.

The sample was prepared using SISPROT. The workflow of TMT-based quantitative proteomics analysis consisted of seven key steps. Firstly, 50 µl of protein sample was mixed with DTT and vortexed. The mixture was incubated at 37 °C for 30 min and allowed to cool to room temperature. IAA was then added and the reaction was carried out in the dark at room temperature for 30 min. Additional DTT was added and the reaction continued at room temperature for 15 min. Secondly, two hydrophilic carboxylated magnetic beads, mixed in a 1:1 ratio, were incubated at 1,000 rpm and 24 °C for 10 min. The protein was precipitated and the magnetic bead fraction was separated using magnetic isolation. Thirdly, the magnetic beads were resuspended in EPPS buffer, LysC and Trypsin were added and the mixture was incubated at 37 °C with rotation for 16 h. Fourthly, after enzymatic digestion, the samples were transferred to a new tube and resuspended in EPPS buffer (pH 8.5). Fifthly, TMT18plex labeling was performed as indicated in the table below, with the corresponding samples. The reaction was carried out at room temperature for 1 h. Sixthly, following the reaction, 5% hydroxylamine solution was added and the mixture was incubated at room temperature for 15 min to terminate the reaction. The labeled samples were combined in equal amounts and desalted at 45 °C. Lastly, after drying, the samples were subjected to HPLC fractionation, resulting in 48 fractions, which were then combined into 24 fractions. The resulting peptide samples were dried and analyzed.

For liquid chromatography–tandem mass spectrometry (LC–MS/MS) analysis, the peptide samples were resuspended in a solution of 0.1% (v/v) formic acid in water. LC was performed using a low-pH mobile phase separation on an Easy-nLC 1200 system coupled to a Q-Exactive HF-X Orbitrap MS instrument (Thermo Fisher Scientific). For LC, mobile phase A was 0.1% (v/v) formic acid in water and mobile phase B was 0.1% formic acid in acetonitrile. The column used was a reverse-phase analytical column (25 cm, inner diameter: 75 µm). The gradient settings were as follows: 0 min, 3% B; 5 s, 5% B; 24 min, 15% B; 45 min, 28% B; 52 min 30 s, 38% B; 52 min 35 s, 100% B; 65 min, 100% B; stop. The flow rate was maintained at 300 nl min^{−1}. MS was performed in positive ion mode using the Orbitrap mass analyzer. Full-scan MS spectra were acquired in the range of 350–1,500 Da with a resolution of 60,000 (profile mode). Data-dependent acquisition was used for MS/MS, with centroid mode for the spectra. The 15 most intense ion peaks were selected for secondary high-energy collision-induced dissociation fragmentation. MS/MS spectra were acquired at a resolution of 45,000. Additional parameters included an isolation window of 0.7 *m/z*, default charge state of 2+, normalized collision energy of 35%, maximum injection time of 50 ms and dynamic exclusion set to 30.0 s.

For data analysis, the raw data were processed using Proteome Discoverer (version 2.5, Thermo Fisher Scientific). The analysis was performed against the human UniProt FASTA database with an integration tolerance of 20 ppm. Carbamidomethylation was specified as a fixed modification for fragment ions. A false discovery rate of 1% was applied at the peptide spectrum match and peptide levels. Statistical significance was determined using a *t*-test.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main article and Supplementary Information. Source data are provided with this paper.

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2023YFA1506500 to P.R.C.; 2021YFA0910800 and 2023YFA1506500 to J.W.), Beijing Natural Science Foundation (Z200010 to P.R.C.), New Cornerstone Science Foundation through the New Cornerstone Investigator Program (to P.R.C.), Young Elite Scientists Sponsorship Program from China Association for Science and Technology (YESS20200010 to J.W.), Shenzhen Innovation of Science and Technology Commission Grant (JCYJ20210324104210028, RCYX20210609103118010 and QTD20221101093558015 to J.W.) and Shenzhen Medical Research Funding (A2303067 to J.W.).

Author contributions

P.R.C. and J.W. designed the research and wrote the paper with input from all authors. Z.G., T.L., H.Y., W.S.C.N. and H.W. conducted most of the experiments.

Competing interests

The authors declare no competing interests.

Additional information

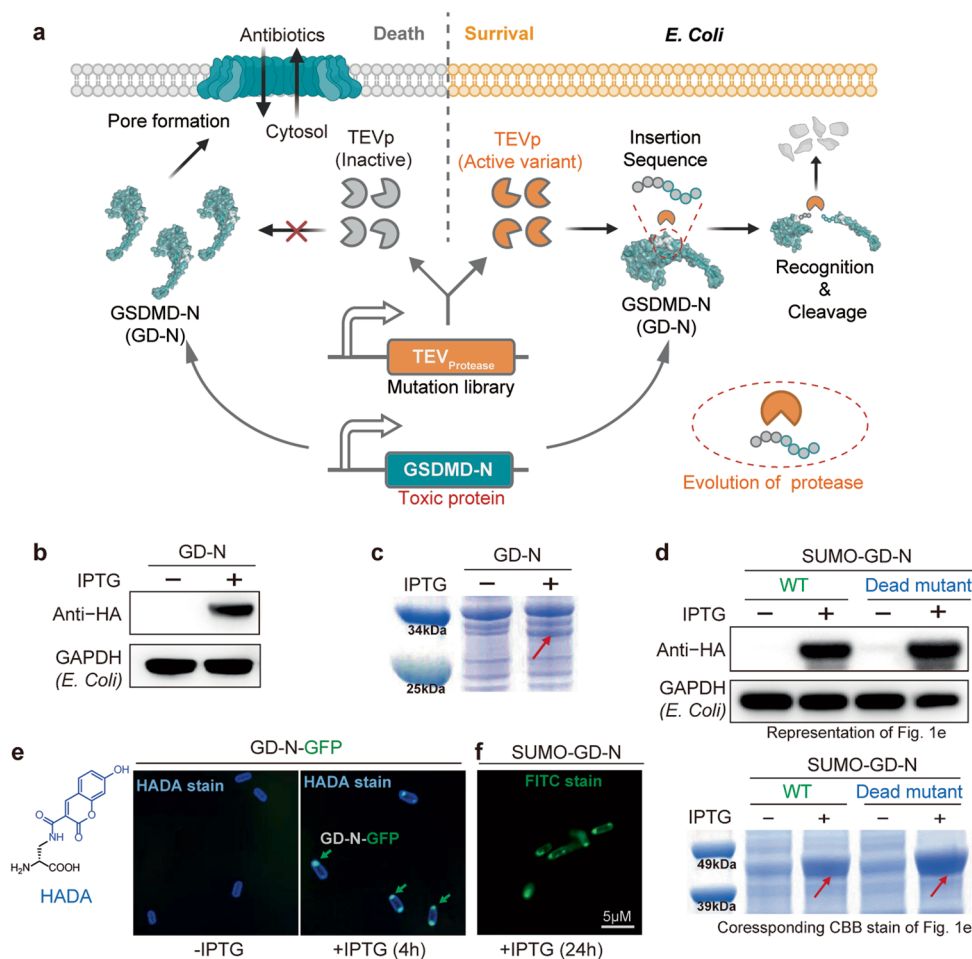
Extended data is available for this paper at <https://doi.org/10.1038/s41589-025-02063-3>.

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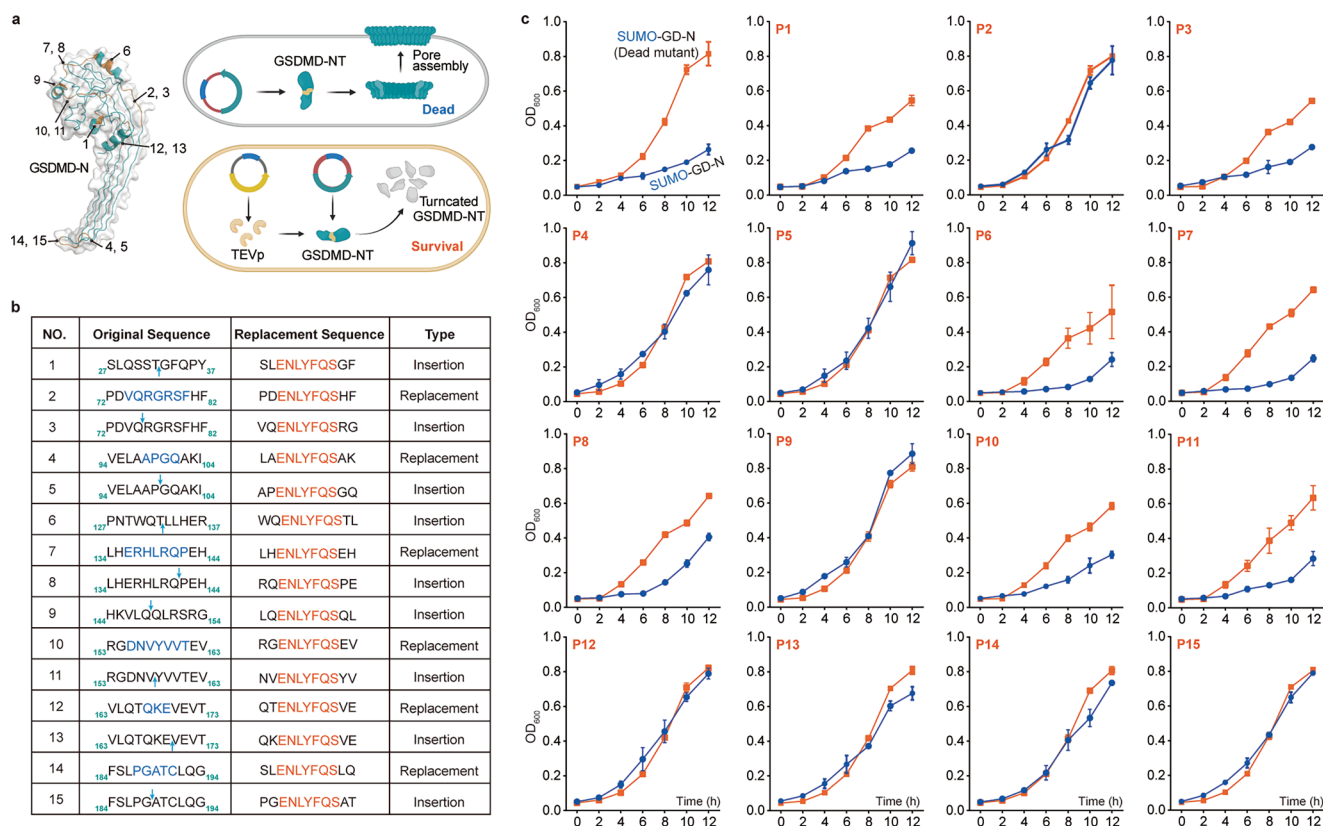
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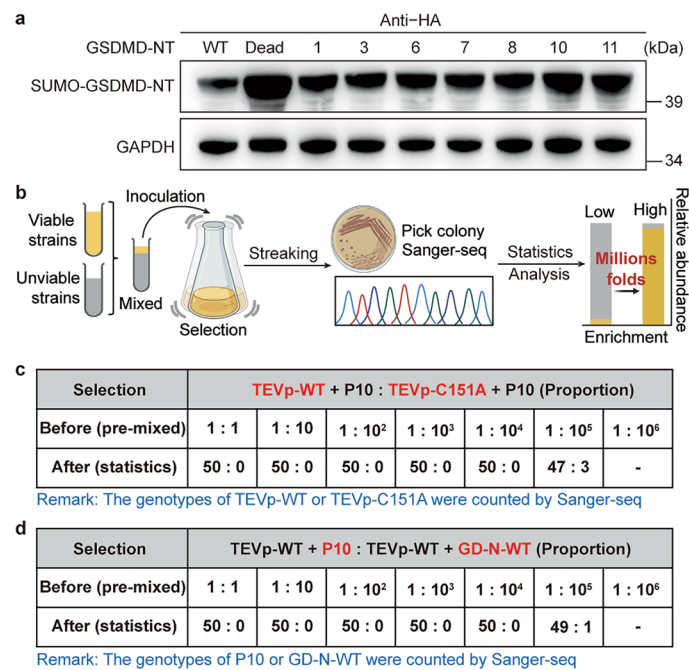
Extended Data Fig. 1 | Expression and function of the GSDMD-N protein in *E. coli*. **a**, Schematic mechanism of the protease reprogramming process utilizing the Gasdermin D-based selection system. A specific sequence for recognition was incorporated into a flexible loop of a toxic protein (GSDMD-N, GD-N) to establish an in vivo selection system. If a protease variant successfully cleaves the targeted sequence, GD-N becomes non-toxic, allowing the host to survive. If not, GD-N could be assembled into a pore structure on bacteria membrane, leading to the death of the host bacteria. This system coupled the protease activity and cell growth with high fidelity and robustness, which can be further utilized to evolve the activity and specificity of the desired protease. **b, c**, Expression of the GD-N

protein in *E. coli*. **d**, Expression of the SUMO-GD-N protein in *E. coli*. **d** reproduced the data shown in Fig. 1e and was included here to facilitate direct comparison with coomassie brilliant blue (CBB) stain result. **e**, Location of expressed GD-N-GFP in *E. coli*. The cell wall of *E. coli* was first labeled with HADA for 4 hours, after which GD-N-GFP was expressed for 6 hours. The GD-N-GFP accumulated on the membrane, but the membrane structure had not yet to be destroyed. **f**, Membrane staining of *E. coli* after the expression of SUMO-GSDMD-N for 24 h. The membrane of *E. coli* was stained with FITC for 1 h. Schematic diagrams were created with [BioRender.com](https://www.biorender.com).

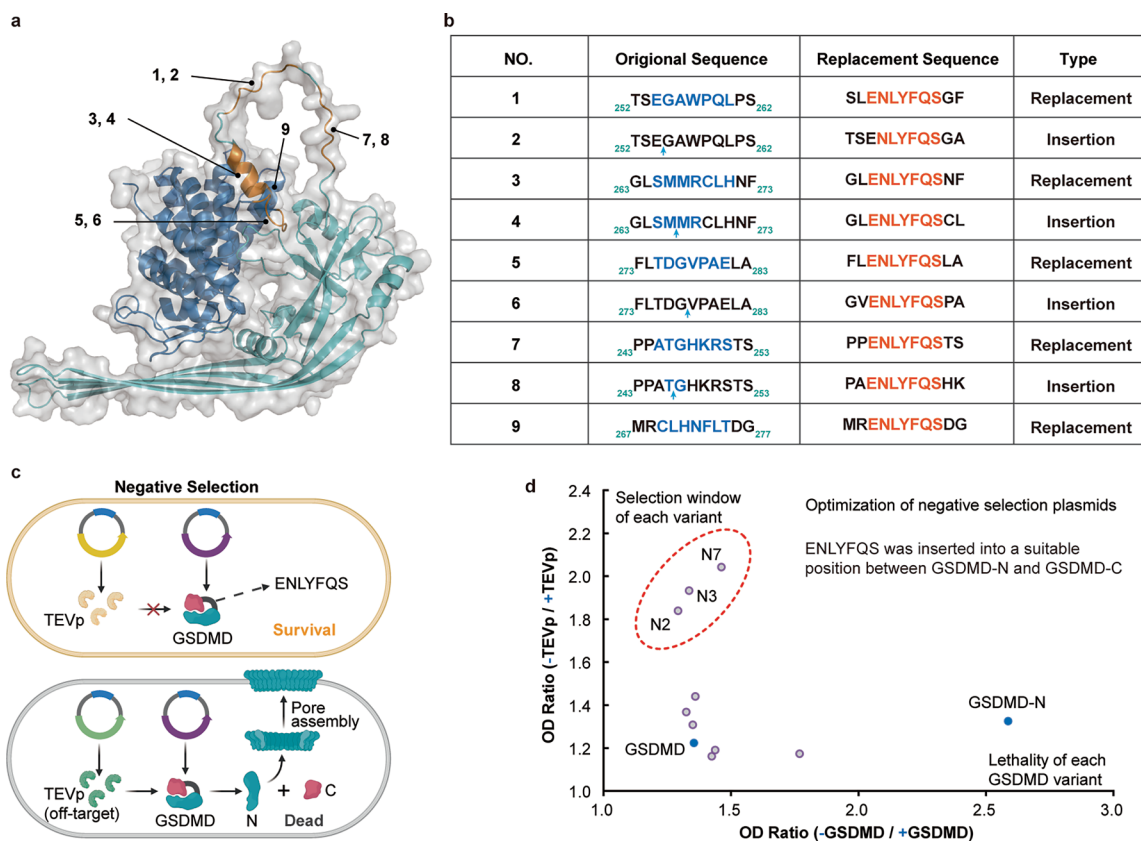


Extended Data Fig. 2 | Screening of optimal selection plasmid. a, Crystal structure showed the insertion positions of the target sequences on SUMO-GD-N protein (PDB: 6VFE), and the different insertion positions would determine whether this modified SUMO-GD-N version met the requirements (as showed in the schematic diagram). **b**, The detailed amino acid sequence information of various SUMO-GD-N versions (P1-P15). **c**, The toxic effect of different SUMO-

GD-N versions (P1-P15) in *E. coli*. The SUMO-GD-N (Dead mutant) version which hardly affects bacterial growth was used as a positive reference, and the wild-type SUMO-GD-N which strongly suppressed bacterial growth was used as a negative reference. This experiment was performed three times. Schematic diagrams were created with [BioRender.com](https://www.biorender.com).

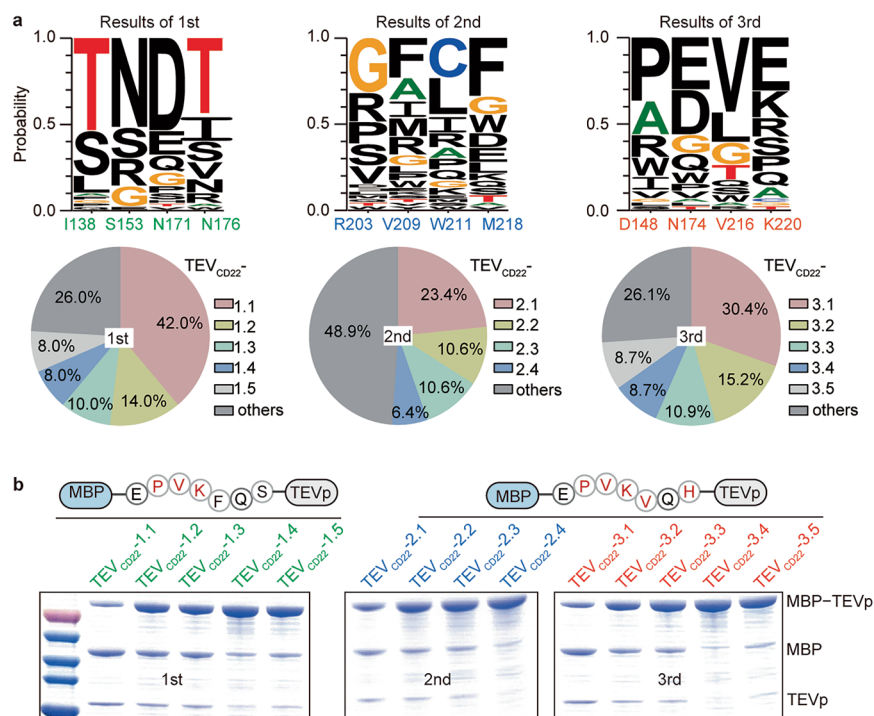


Extended Data Fig. 3 | Validation of the optimal reprogramming system. a, Western blot detected the expression of different versions of SUMO-GD-N. **b,** Schematic diagram of the in vivo selection process. **c,** Proportion of TEVp-WT / TEVp-C151A genotype before and after the in vivo selection process. **d,** Proportion of P10 / GD-N-WT genotype before and after the in vivo selection process. Schematic diagrams were created with [BioRender.com](#).



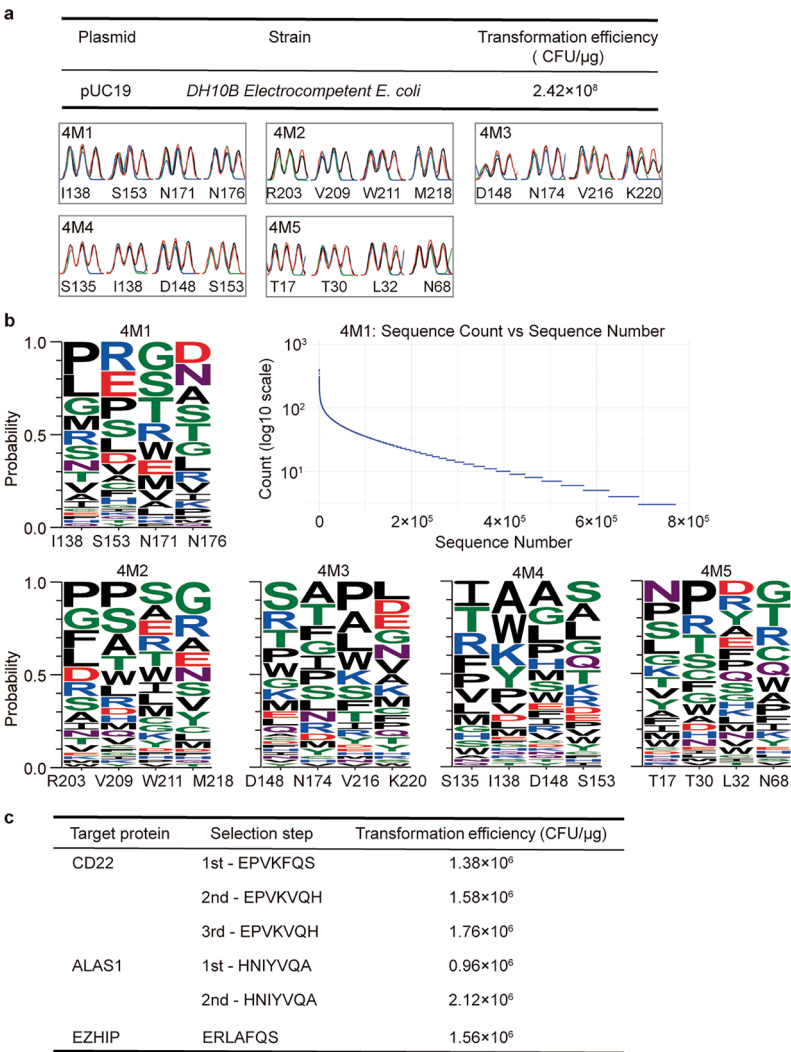
Extended Data Fig. 4 | Design and functional evaluation of the negative selection system for protease evolution. a, Predicted structure of GSDMD protein (AF-P57764-F1). **b**, The detailed amino acid sequence information of various GD-F versions (N1-N9). **c**, Schematic diagram of negative selection system. **d**, Evaluation of the N1-N9 variants to identify suitable SUMO-GSDMD variants for further applications. The x-axis was the OD₆₀₀ ratio between bacteria

without the SUMO-GSDMD protein and bacteria expressing SUMO-GSDMD, representing the toxicity of each SUMO-GSDMD variant; the y-axis was the OD₆₀₀ ratio between bacteria without TEV expression and bacteria expressing the TEV protease, representing the ability of TEV protease to cleave the ENLYFQS sequence and to suppress bacterial growth. Schematic diagrams were created with [BioRender.com](https://www.biorender.com).



Extended Data Fig. 5 | Evolution of the TEV protease for the target cleavage of CD22 protein. a, Enriched mutation pattern of TEV protease after the 1st, 2nd and 3rd rounds of selection at different mutation sites. Pie charts indicated the

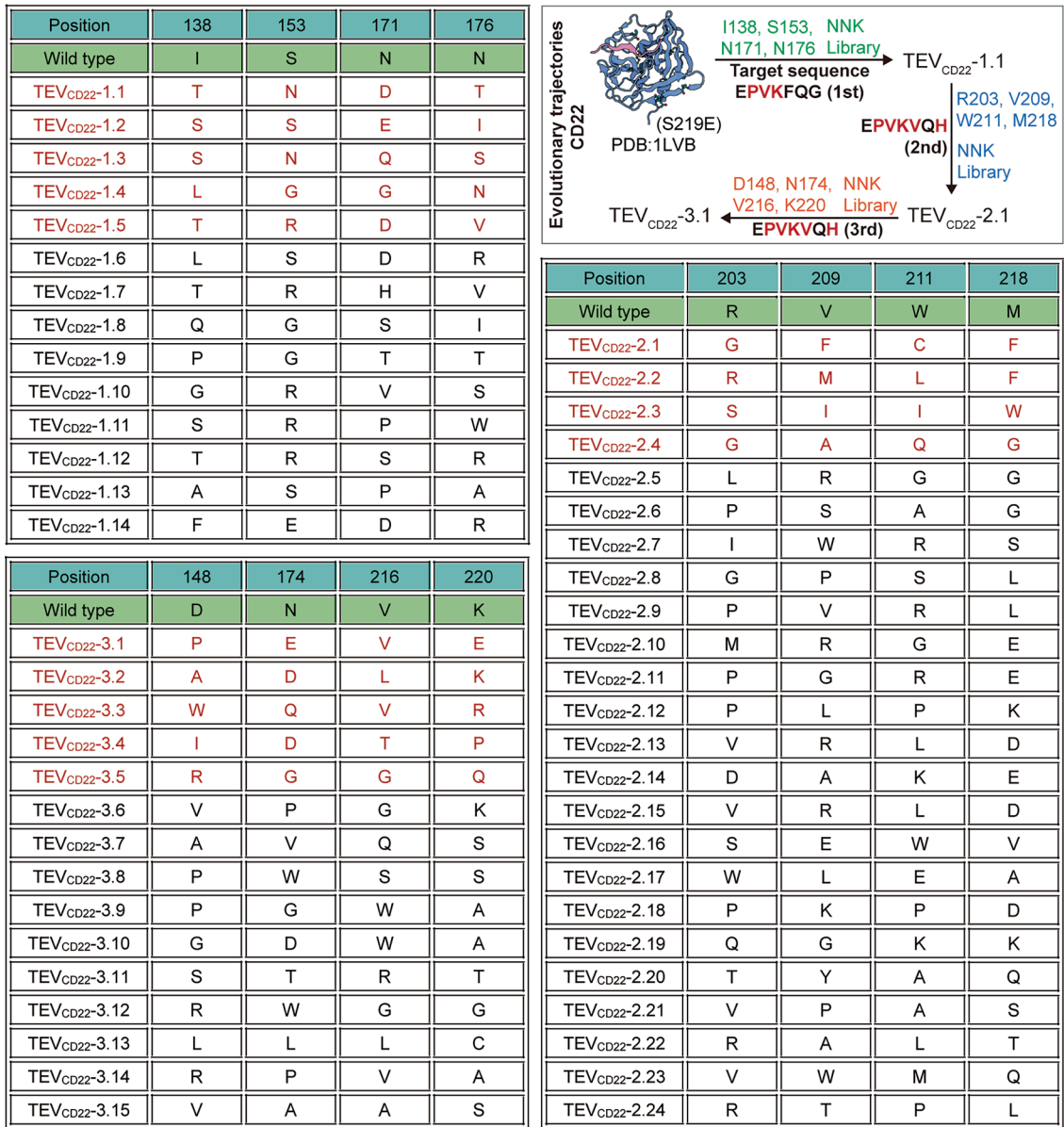
proportion of different TEV variants in selection progress. **b**, Comparison of evolved TEV variants in 3 rounds selection progress for the cleavage of the target sequence.



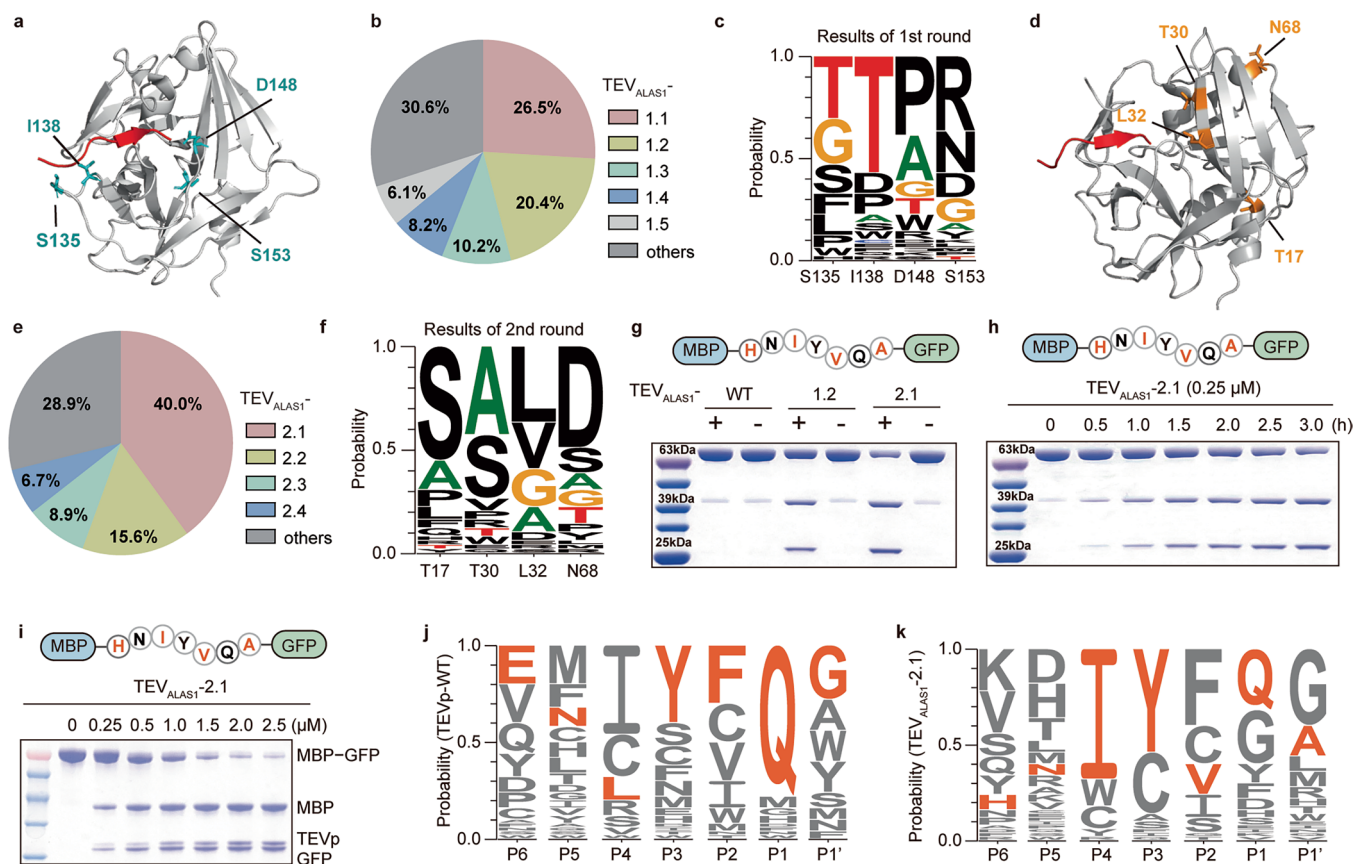
*Strain: BL21(DE3) Competent *E. coli* (harbor a selection plasmid)

Extended Data Fig. 6 | The complexity of mutant-libraries and the transformation efficiency in the individual selection rounds. a, the transformation efficiency of the competent cells used for constructing the mutant library in this work (with the standard plasmid pUC19) and the chromatograms of the mutated sites. **b**, The WebLogo representing amino acid

diversity was generated using an online tool (<https://weblogo.threeplusone.com/>). For further analysis, the 4M1 library was selected for next-generation sequencing (NGS). **c**, the transformation efficiency achieved in each individual selection.



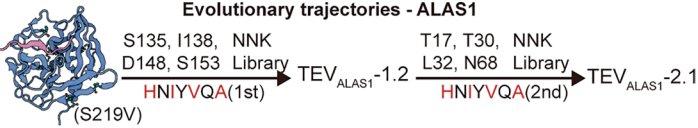
Extended Data Fig. 7 | Schematic diagram showed the evolutionary pathway of selection progress. Lists showed the sequence information of variants in CD22-1st, -2nd and -3rd round selection progresses.



Extended Data Fig. 8 | Evolution of the TEV protease for the target cleavage of ALAS1 proteins. **a**, Structure and mutation sites of the TEV protease in the round-1 evolution of TEVp (PDB: 1LVM). **b**, Pie chart indicated the proportion of different TEV variants in ALAS1-1st round selection progress. **c**, Enriched mutation pattern of TEVp after the first round of selection. **d**, Structure and mutation sites of the TEV protease in the round-2 evolution of TEVp (PDB: 1LVM). **e**, Pie chart indicated the proportion of different TEV variants in ALAS1-2nd round selection progress. **f**, Enriched mutation pattern of TEVp after the second round of selection. **g**, Comparison of evolved TEV variants in ALAS1-1st round selection progress for the cleavage of the target sequence. The MBP-HNIYVQA-GFP

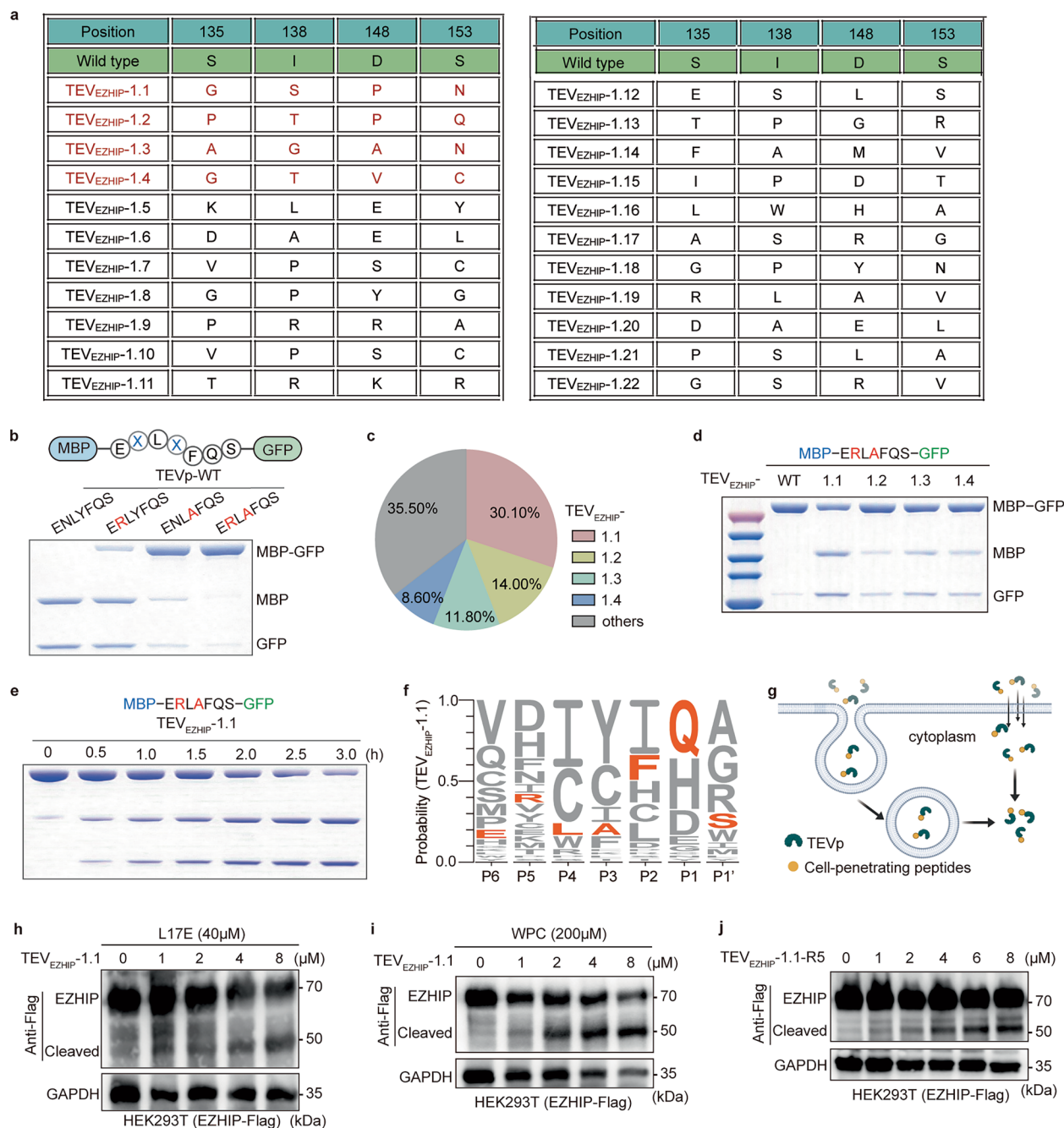
was utilized as the substrate, and TEVp-WT was used as the control. **h**, The cleavage kinetics of TEV_{ALAS1}⁻2.1 to MBP-HNIYVQA-GFP. **i**, Cleavage rate of MBP-HNIYVQA-GFP by different concentration of TEV_{ALAS1}⁻2.1. The concentration of MBP-HNIYVQA-GFP is 25 mM. **j-k**, Based on this evolution system, sequence tolerance at each position of the recognition motif was systematically evaluated for both TEVp-WT and TEV_{ALAS1}⁻2.1. By randomizing each site of the substrate and then conducting the selection process, the information of sequence tolerance at each site was obtained. The substrate recognition patterns differed substantially between the wild-type and evolved variants, indicating that the specificity of TEVp was reprogrammed rather than evolved to be more promiscuous.

Position	135	138	148	153
Wild type	S	I	D	S
TEV _{ALAS1} -1.1	T	T	P	R
TEV _{ALAS1} -1.2	G	T	A	N
TEV _{ALAS1} -1.3	F	D	P	D
TEV _{ALAS1} -1.4	S	P	W	G
TEV _{ALAS1} -1.5	L	T	T	R
TEV _{ALAS1} -1.6	W	W	T	G
TEV _{ALAS1} -1.7	P	F	K	T
TEV _{ALAS1} -1.8	S	C	R	K
TEV _{ALAS1} -1.9	S	S	D	L
TEV _{ALAS1} -1.10	G	P	S	P
TEV _{ALAS1} -1.11	W	S	I	Y
TEV _{ALAS1} -1.12	S	A	A	A
TEV _{ALAS1} -1.13	L	T	G	D
TEV _{ALAS1} -1.14	P	A	G	Y
TEV _{ALAS1} -1.15	L	Q	E	M
TEV _{ALAS1} -1.16	T	E	G	A
TEV _{ALAS1} -1.17	H	R	R	R
TEV _{ALAS1} -1.18	P	W	G	G



Position	17	30	32	68
Wild type	T	T	L	N
TEV _{ALAS1} -2.1	S	A	L	D
TEV _{ALAS1} -2.2	S	S	V	D
TEV _{ALAS1} -2.3	A	S	V	T
TEV _{ALAS1} -2.4	A	V	A	D
TEV _{ALAS1} -2.5	F	S	D	G
TEV _{ALAS1} -2.6	P	W	G	G
TEV _{ALAS1} -2.7	R	P	S	A
TEV _{ALAS1} -2.8	S	A	A	A
TEV _{ALAS1} -2.9	L	T	G	D
TEV _{ALAS1} -2.10	P	A	G	Y
TEV _{ALAS1} -2.11	L	Q	E	M
TEV _{ALAS1} -2.12	T	E	G	A
TEV _{ALAS1} -2.13	H	R	R	R
TEV _{ALAS1} -2.14	Q	S	A	P
TEV _{ALAS1} -2.15	S	P	G	L

Extended Data Fig. 9 | Schematic diagram showed the evolutionary pathway of selection progress. Lists showed the sequence information of variants in ALAS1-1st and -2nd round selection progress.



Extended Data Fig. 10 | Evolution of the TEV protease for target cleavage of EZHIP proteins and Cell-penetrating peptide screening. **a**, The sequence information of variants in EZHIP-selection progress. **b**, Comparison of cutting efficiency TEVp-WT for the target sequence and intermediate sequence by using MBP-EXLXFQS-GFP as the substrate. **c**, Pie chart indicated the proportion of different TEV variants in EZHIP-selection progress. **d**, Comparison of cutting efficiency of evolved TEVp hits for the MBP-ERLAFQS-GFP substrate. **e**, The cleavage kinetics of TEV_{ALAS1}-2.1 to MBP-ERLAFQS-GFP. **f**, Based on this evolution system, sequence tolerance at each position of the recognition motif was

systematically evaluated for TEV_{EZH1P}-1.1. **g**, Schematic diagram of cell-penetrating peptide delivering TEV_{EZH1P}-1.1 into cytoplasm. **h**, L17E mixed with different concentration of TEV_{EZH1P}-1.1, then HEK-293T cells transfected with EZHIP were treated with the mixture. **i**, WPC mixed with different concentration of TEV_{EZH1P}-1.1, then HEK-293T cells transfected with EZHIP were treated with the mixture. **j**, HEK-293T cells transfected with EZHIP were treated with different concentration of TEV_{EZH1P}-1.1-R5 (TEV_{EZH1P}-1.1 fused with 5 Arg). Using western blot to detect the cleavage of variant on EZHIP. This experiment was performed three times with similar results. Schematic diagrams were created with [BioRender.com](https://www.biorender.com).

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- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
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- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
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Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

The fluorescence intensity of FRET and OD600 were collected by microplate reader (Tecan). The microscopic images were captured with an inverted fluorescence microscope (DMI4000B, Leica) or LSM 900 confocal microscope (Zeiss). The images of western blots and gels were captured with an automatic chemiluminescence image analysis system (Tanon Chemi Dog Ultra). LC-MS/MS analysis was carried out with AB Sciex QTrap 6500 plus mass spectrometer and UPLC system (1290, Agilent).

Data analysis

We used the ImageJ to quantify the protein bands, and used the Graph Pad Prism Software 8.0 to generate the graph figures and perform statistical analyses. Raw data files generated by LC-MS/MS were processed using Proteome Discoverer (version 2.5, Thermo Fisher Scientific). The analysis was performed against the human UniProt FASTA database (UniProt human-filtered-reviewed), with intergration tolerance of 20 ppm.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Structures of GSDMD-N and predicted GSDMD for modification were obtained from the RCSB Protein Data Bank with accession codes 6VFE and AF-P57764-F1. Structures of TEV protease for evolution were obtained from the RCSB Protein Data Bank with accession codes 1LVM. Structures of CD22 were obtained from the RCSB Protein Data Bank with accession codes 5VKJ and 7O52.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Not applicable.
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable.
Population characteristics	Not applicable.
Recruitment	Not applicable.
Ethics oversight	Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size was determined to be adequate based on the magnitude and consistency of measurable differences between groups. Sample sizes were chosen as large as possible while taking into account the experimental effort required to generate the respective data.
Data exclusions	No data were excluded from the analyses.
Replication	All experimental findings were reliably reproduced. Multiple independent repeats were included for related experiments.
Randomization	No randomization was used in this study, as the experiments were quantitative in vitro biochemical assays.
Blinding	No blinding was used in this study, as objective, quantitative biochemical experiments were performed and samples were treated identically.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	For western blotting: anti-HA antibody (YM3003, Immunoway,1:3000), anti-His antibody (YM3004, Immunoway,1:3000), anti-Flag antibody (F1804, Merck,1:3000), anti-GAPDH antibody (ab8245, Abcam,1:3000), anti-ALAS1 antibody (A6521, Abclonal,1:2000), anti-EZH1P antibody (ab313392, Abcam,1:2000), anti-H3K27me3 antibody (39055, Active Motif,1:2000), anti-H3K27ac antibody (39133, Active Motif,1:2000), anti-H3 antibody (17168-1-AP, Proteintech,1:2000), HRP-conjugated Goat anti-Mouse secondary antibody (RS0001, Immunoway,1:5000), HRP-conjugated Goat anti-Rabbit secondary antibody (RS0002, Immunoway,1:5000).
Validation	All antibody used in our study have been validated by the manufacturers and detailed information could be found on the website from manufactures as listed below. anti-HA antibody (YM3003), https://www.immunoway.com.cn/products/primary-antibodies/YM3003-HA-Tag-1B10-Mouse-mAb.html ; anti-His antibody (YM3004), https://www.immunoway.com.cn/products/primary-antibodies/YM3004-His-Tag-4E6-Mouse-mAb.html ; anti-Flag antibody (F1804), https://www.sigmaaldrich.cn/CN/zh/product/sigma/f1804 ; anti-GAPDH antibody (ab8245), https://www.abcam.cn/products/primary-antibodies/gapdh-antibody-6c5-loading-control-ab8245.html ; anti-ALAS1 antibody (A6521), https://abclonal.com.cn/catalog/A6521 ; anti-EZH1P antibody (ab313392), https://www.abcam.cn/products/primary-antibodies/ezhip-antibody-epr28396-50-ab313392.html ; anti-H3K27me3 antibody (39055), https://www.activemotif.com.cn/catalog/details/39155/histone-h3-trimethyl-lys27-antibody-pab ; anti-H3K27ac antibody (39133), https://www.activemotif.com.cn/catalog/details/39133/histone-h3-acetyl-lys27-antibody-pab ; anti-H3 antibody (17168-1-AP), https://www.ptgcn.com/products/Histone-H3-Antibody-17168-1-AP.htm ; HRP-conjugated Goat anti-Mouse secondary antibody (RS0001), https://www.immunoway.com.cn/products/secondary-antibodies/RS0001-Goat-Anti-Mouse-IgG-H-L-HRP.html ; HRP-conjugated Goat anti-Rabbit secondary antibody (RS0002), https://www.immunoway.com.cn/products/secondary-antibodies/RS0002-Goat-Anti-Rabbit-IgG-H-L-HRP.html

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	293T cells, U2OS cells and HepG2 cells were purchased from National Biomedical Cell-line Resource Center, China.
Authentication	All cell lines used in this study were authenticated by STR profile report.
Mycoplasma contamination	All cell lines in our laboratory were routinely tested for mycoplasma contamination and cells used in this study were mycoplasma free.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>